

Monetary Policy and Earnings Inequality: Evidence from High-Frequency Administrative Data*

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Abstract

We study how monetary policy affects earnings inequality using new high-frequency administrative tax data from eurozone member Estonia. Monetary policy shocks substantially affect earnings inequality, with most of the effect originating from the extensive margin, that is, transitions into and out of employment. Combining our dataset with granular MPC estimates, we show that the earnings heterogeneity channel amplifies the aggregate MPC and consumption response, accounting for 7.4% of the response. Regarding the recent inflation surge, we find that monetary policy shocks have a stronger overall impact in high-inflation periods, driven largely by labour market exits, which further amplifies aggregate transmission.

Keywords: Monetary policy, labour income inequality, inflation, state dependency, earnings heterogeneity channel, aggregate MPC

JEL classification: E52, D31, J31, J63

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1 Introduction

Monetary policy shocks may have important implications for inequality, which, in turn, can amplify their aggregate transmission. Specifically, the earnings heterogeneity channel of monetary policy (see e.g., Auclert 2019) captures that monetary policy affects workers unequally, with disproportionate earnings losses for some workers. However, to what extent this channel and the aggregate amplification are reflected in the data remains an important empirical question.¹ Against this background, we exploit near-real-time high-frequency administrative data from Estonia, a euro area member state, to address these questions, focusing on earnings heterogeneity. Importantly, the high frequency of our data allows us to cleanly separate whether the adjustment is driven by earnings changes among continuing workers, that is, the intensive margin, or by transitions into and out of employment, that is, the extensive margin.

We estimate the distributional effects of monetary policy across the earnings distribution using monthly data, showing that low-income households are more affected. Exploiting that we can disentangle the total effect in the extensive and intensive margin, we show that the extensive margin, that is, entry and exit in the labour market, mostly explains the heterogeneous response. We then use our estimates to evaluate how the earnings heterogeneity channel of monetary policy affects the aggregate dynamics, in particular, the response of aggregate consumption. We complement our administrative data with household-level marginal propensity to consume (MPC) estimates. We can use this combined dataset to calculate the aggregate MPC for a monetary policy shock and back out the role of the earnings heterogeneity channel, inspired by the approach of Patterson (2023). Our estimate assigns 7.4% of the impact of monetary policy shock on aggregate consumption to the earnings heterogeneity channel, with the extensive margin being quantitatively more important. Finally, we show that the effects for heterogeneity and amplification are somewhat stronger during periods of high inflation, exploiting that the high frequency of our dataset enables us to analyse influential but short-lived episodes such as the recent inflation surge.

The challenge in assessing the impact of monetary policy on earnings inequality and its subsequent role in amplifying (or dampening) the shock itself is the access to sufficient granular data that also features information on individual MPCs. For this reason, we use individual-level administrative tax data from the euro area member Estonia to exploit

¹For instance, recent evidence based on Norwegian data from Bilbiie et al. (2025) points to the limited role, or even near-irrelevance, of aggregate demand amplification through heterogeneity.

several key features. The first is that wage and employment data are available for the entire population without top coding at a monthly frequency. Therefore, we can measure the distributional implications of monetary policy shocks for the entire earnings distribution. Due to the monthly frequency, we can even distinguish between the extensive and intensive margins. The data also allows for evaluating state dependencies for short-lived but influential periods such as the recent inflation surge. A second advantage is that the data are available almost in real time, covering the period from January 2005 to December 2025. Thus, the dataset includes recent events. Third, Estonia joined the euro area in 2011 and had a fixed exchange rate against the euro before that. Therefore, we can study the monetary policy shocks of the European Central Bank (ECB). We use a high-frequency approach that captures the changes in interest rates in a tight window around meetings of the Governing Council (Jarociński and Karadi 2020). Fourth, we can combine the administrative tax data with survey data from the Household Finance and Consumption Survey (HFCS) on the marginal propensity to consume. This combined dataset disentangles how the distributional impact of monetary policy affects aggregate consumption. Fifth, the administrative data and HFCS data is available to other researchers.²

We first document that a contractionary monetary policy shock has a heterogeneous negative effect on earnings.³ In particular, the impact is stronger for low-wage earners, as they respond particularly strongly to monetary policy. While the average household's labour income reacts less strongly and especially high-income earners are almost not affected, we observe a slight increase in wage income for the top earners (0.1%). In that regard, our response features a weak U-shape. Using our granular data, we decompose the overall effect into the intensive margin of wage changes and the extensive margin of labour market entries and exits. The results for the low-income group are driven by a stronger entry and exit response. The intensive margin plays a relatively larger role for households in the middle and at the top of the income distribution. To sum up, we find that the earnings heterogeneity channel is present in our data. Additionally, the households' exposure to the intensive and extensive margins is also heterogeneous.

While individual-level administrative data features the granularity to assess the potential impact of monetary policy on wage inequality, such data is often only available

²The administrative data can be accessed from Statistics Estonia, as explained on their website. The HFCS data can be accessed from the ECB, as described on their Household Finance and Consumption Network webpage.

³Monetary policy may also affect households differentially through other channels, e.g. income composition channel, portfolio channel, and savings redistribution channel, which are not assessed in our analysis.

or used at an annual frequency. Our data highlights that annual data overestimates the relative importance of the intensive margin, as yearly data misses substantial within-year transitions into and out of employment. For instance, we find that the intensive margin accounts for one-third of the total effect in monthly data, but for two-thirds in yearly data. Thus, we exploit the fact that our data captures shorter spells of unemployment more accurately, thereby making a cleaner distinction between the intensive and extensive margins.

The next step is to evaluate the aggregate implications of the distributional effects of monetary policy. To what extent does the earnings heterogeneity channel amplify aggregate dynamics, and is this amplification driven by the intensive or extensive margin? Specifically, we measure how the distributional effects of monetary policy shape the aggregate consumption response, building on the matching-multiplier approach of Patterson (2023). This requires matching individuals' exposure to the monetary policy shock with estimates of their individual marginal propensities to consume (MPCs). Our Estonian data are particularly useful for this purpose. We complement our administrative data with household-level MPC estimates from the Household Finance and Consumption Survey for Estonia in 2021. We then match surveyed households to their members' exposure to the monetary policy shock in our administrative data.

This approach provides an aggregate MPC for monetary policy shocks, which can be decomposed into two components: the income-weighted average MPC and the covariance between the relative household-level earnings response to monetary policy and the MPC. If exposure to the shock and the MPC are positively correlated, the earnings heterogeneity channel amplifies the response of aggregate consumption to a monetary policy shock. Using a microsimulation approach to calculate the aggregate MPC for monetary policy shocks, we find that 7.4% of the aggregate response stems from the heterogeneous earnings responses of workers to the monetary policy shock and can therefore be attributed to the earnings heterogeneity channel. This reflects the fact that low-income households have high MPCs and are also particularly affected by monetary policy shocks. Furthermore, the monetary policy shock increases earnings inequality, as measured by the Gini coefficient using our granular data.

We also distinguish between the roles of the intensive and extensive margins in shaping the aggregate MPC. In particular, we find that the aggregate MPC for a monetary policy shock implies a smaller role for earnings heterogeneity when we focus only on the intensive margin, while the amplification is stronger when we focus on the extensive margin. Overall, both margins amplify the response to monetary policy shocks, but the amplification

operates more strongly through labour market entry and exit.

One important advantage of our data is its near-real-time availability and high frequency, especially relative to annual or infrequently updated data. These features allow us to study relatively short-lived but influential episodes. In particular, state dependence and nonlinearities may shape the transmission of monetary policy during specific periods.⁴ Motivated by the recent inflation surge, we analyse how the inflation environment affects the earnings heterogeneity channel of monetary policy and its aggregate amplification.⁵

We find that the inflation environment affects the distributional impact of monetary policy. Specifically, the response of total earnings is considerably more negative across almost the entire earnings distribution in the high-inflation regime. Additionally, low-wage workers are affected substantially more strongly than middle- or high-wage workers in a high-inflation environment. The inflation regime also affects the margins of adjustment. The lowest earnings deciles are more exposed to the monetary policy shock in the high-inflation regime, primarily through the extensive margin. On average, the extensive margin accounts for slightly less than half of the response in the low-inflation regime and for slightly less than 80% in the high-inflation regime.

Furthermore, we show that the aggregate MPC is higher in the high-inflation regime due to a stronger earnings heterogeneity channel. However, focusing only on the aggregate MPC does not account for the overall size of the shock’s effect. Monetary policy shocks have a much larger quantitative impact when inflation is high. Therefore, the total aggregate effect is considerably more pronounced in the high-inflation regime, driven by both a stronger overall response and stronger heterogeneous exposure.

An important consideration for our findings is their external validity. One advantage is that Estonia is part of the euro area, and therefore, we can study the monetary shocks of the ECB. Importantly, our baseline findings, for which we do not condition on inflation or the size of shocks, are in line with similar studies that were conducted in Denmark, Norway, Sweden, Germany and France (Andersen et al. 2023, Holm et al. 2021, Amberg et al. 2022, Broer et al. 2022, Hubert and Savignac 2024). Furthermore, we also show that the aggregate risk exposure of individual earnings to GDP is very similar to the US, as studied in Guvenen et al. (2017). Additionally, the labour market features high flexibility, being much closer to the US than to Western Europe or Scandinavia. However, Estonia

⁴Gargiulo et al. (2024) find that the effects of monetary policy depend on the level of inflation using aggregate data.

⁵We want to emphasise that our sample contains only two periods of very elevated inflation, namely a period around the financial crisis and the period following the pandemic.

has been growing relatively strongly during this period, starting from a substantially lower economic level than the US or the average in the euro area. Furthermore, Estonia has a flat tax rate scheme, which could affect the incentives in the labour market.

Related literature This paper is part of the literature that studies the distributional impact of monetary policy using survey and administrative data. Coibion et al. (2017) find from survey data that monetary policy increases inequality in labour earnings in the US. Andersen et al. (2023), Holm et al. (2021), Amberg et al. (2022) use detailed yearly administrative data from Scandinavia to estimate the impact of monetary policy on inequality. Overall, these studies suggest a stronger earnings response for low-income households. Broer et al. (2022) estimate the distributional impact of monetary policy using monthly (top-coded) employment data from 1995 to 2013 for Germany. They argue that the extensive margin is very relevant for low-income workers. Hubert and Savignac (2024) use annual French matched administrative-survey data to evaluate the roles of the extensive and intensive margins. Our contribution to studying the impact is threefold. First, our results confirm previous empirical findings of these studies using a different country. Second, we use a high-quality administrative dataset without top coding from a euro area country that has not yet been used in this context. Third, we show that monthly data provide a cleaner estimation of the intensive and extensive margins as they account properly for the labour market spell of workers.

However, our study differs from previous work in one key dimension. We analyse the role of the earnings heterogeneity channel in the aggregate amplification of monetary policy shocks, an area for which direct empirical evidence remains limited.⁶ Recent work on matching multipliers by Patterson (2023) shows that heterogeneous exposure of households can amplify the aggregate consumption response. More recently, Bilbiie et al. (2025) estimate the aggregate MPC using Norwegian data. While their evidence points to limited, or even negligible, aggregate amplification through heterogeneity when focusing on total income, they find some amplification when focusing specifically on earnings heterogeneity. Using a different dataset and empirical strategy, our approach also identifies positive aggregate amplification through the earnings heterogeneity channel. In addition, we distinguish between the roles of the extensive and intensive margins, highlighting the

⁶This directly relates to the literature on HANK models, as coined by Kaplan et al. (2018), and to work on aggregate amplification through heterogeneity. Contributions include Galí et al. (2007), Bilbiie (2008), Bilbiie (2020), Broer et al. (2020), Auclert et al. (2021), Cantore and Freund (2021), Kase et al. (2022), Bayer et al. (2024), Bilbiie (2024), and Debortoli and Galí (2024), among many others.

importance of labour market entry and exit for aggregate amplification.

Our work also complements recent studies that use more indirect approaches combining empirical evidence and theory to quantify consumption effects (e.g., Slacalek et al. 2020, Mäki-Fränki et al. 2022, McKay and Wolf 2023, Lenza and Slacalek 2024, and Pekanov 2024). A key strength of our analysis is that we can match estimated exposure to monetary policy shocks from administrative data with household-level survey evidence on MPCs. More generally, conditional on access to approach administrative earnings data and (survey-based) MPC measures, our approach can be implemented relatively straightforwardly to compute the aggregate MPC for monetary policy shocks in other countries as well.

Furthermore, we relate to the literature on nonlinearities and state dependencies. For instance, Tenreyro and Thwaites (2016) emphasise the importance of state dependence for the monetary transmission of aggregate variables by conditioning the effects of monetary policy shocks on the business cycle. We contribute to this literature by showing that the high-frequency and near-real-time availability of our dataset, which is available to other researchers, allows us to study state dependence in the distributional effects of monetary policy. Gargiulo et al. (2024) use aggregate US data and find that monetary policy transmission depends on the inflation regime, with stronger and more persistent effects when inflation is high. Similarly, Canova and Pérez Forero (2024) show that the transmission of monetary policy shocks to aggregate variables depends on the inflation regime. Using our dataset, we find that the distributional effects of monetary policy also vary with the level of inflation.⁷

2 Data: Labour Income & Monetary Policy

This section first provides a brief institutional background on Estonia before outlining the employed administrative labour income data for the entire population at a monthly frequency. We then discuss the employed high-frequency monetary policy shock series.

2.1 Institutional Context: Labour Market & Monetary Policy

This paper uses data from Estonia, a small open economy that joined the euro area in 2011 and had before a fixed exchange rate with the Euro as an anchor currency as part of the

⁷Our work also relates to recent papers that use quantitative HANK models to study nonlinear dynamics, such as Fernández-Villaverde et al. (2023) and Kase et al. (2022).

European Exchange Rate Mechanism. Importantly, this allows us to study the monetary policy of the ECB. The transmission of monetary policy shocks to the Estonian economy has been documented, in Errit and Uusküla (2014) and Almgren et al. (2022), for instance. The growth rate of loans in Estonia is tightly related to policy interest rates, as there is a high dependency on flexible interest rates that are linked to the 6-month Euribor.

An important institutional feature is the flexible labour market, which is a key margin for absorbing shocks. The labour market is characterised by low nominal wage rigidity and adjustments to shocks through employment status as there is no regular job retention support scheme (Babecký et al. 2010).⁸ The union power is very low, and the minimum wage is the most important labour market institution. Rises in the minimum wage have had a strong effect on wages at the lower part of the wage distribution (Ferraro et al. 2018b) with a limited negative effect on employment (Ferraro et al. 2018a). Taken together, the flexible labour market makes the institutional setting in Estonia, in our opinion, closer to that of the US than to those Western European countries such as France and Germany or the Scandinavian countries. Appendix A provides more details on the aggregate dynamics of the Estonian economy.

2.2 Monthly Administrative Data on Labour Income

Using Estonian administrative tax records gives us monthly and near-real-time data of the earnings distribution. The data on labour income are obtained from the administrative records of the Estonian Tax and Customs Board, which are based on employers' filings. Employers report wages paid to the Tax and Customs Board and all the related taxes of all their employees at a monthly frequency.⁹ The resulting database covers the universe of employees, their total labour income, and their employers. The data are complete since there is no top coding, and the entire set of employees is covered. Importantly, the data is of very high quality as it is used to derive individuals' pension rights at retirement. As a consequence, no additional filtering or data trimming has been imposed.

We construct a database of labour income at the individual level and at a monthly frequency for our horizon from January 2005 to December 2025. To trace entries and exits,

⁸The exception was the Covid-19 crisis when Estonia, like many other EU countries, introduced a temporary job retention support scheme from March 2020 until May 2022. This was an on-off support scheme, which had a temporary effect that muted the reallocation of jobs between firms (Meriküll and Paulus 2024).

⁹These tax declarations are called form TSD, and the employer submits these to the tax office by the 10th day of the month following the payment.

we use a balanced panel of all the individuals who were employed in at least one month in the timespan observed, and we set the labour income equal to zero when the individual was not working. We restrict the age of individuals in our panel to be within the prime working age of 26-65.

The total labour income is derived as the sum of two income sources: i) wage income (from permanent and temporary contracts), and ii) income of employees who serve as executives or board members. While the two types of labour income, wages and remuneration for board members, are very similar in their function and subject to taxation, the executive remuneration is not subject to unemployment insurance payments. Most income comes from wage income; the pay for board members makes only a small contribution, but it could be very important to measure the labour income of the highest income earners. Our labour income covers all wage income, bonuses, and holiday payments and is reported in gross terms.¹⁰ Labour income is deflated to real terms using the Harmonised Index of Consumer Prices (HICP) in 2015 prices.

We sort the individuals into twelve income groups each month by gender and age. The income bins use cutoffs at the 10th, 20th, ..., 90th, 99th, and 99.9th percentiles. The age groups are divided as follows: 26-35, 36-45, 46-55, 56-65.¹¹ We sort into income groups using a proxy of permanent income - the average labour income during the last 12 months.¹² Although it is common to base permanent income measures on a longer history, we explicitly limit the past income to one year only to keep the years around the Great Recession in the sample. Our effective sample in the empirical analysis starts from 2006M2; the income group's derivation requires the first 12 months of data and one month before the monetary policy shock.

The average intensive-margin growth or wage growth of participants is 3.8%, the entry rate is 4.9%, and the exit rate is about 9.3%. We observe that, for most of the distribution, separation rates exceed job-finding rates, which is driven by the fact that we do not capture first-time labour-market entrants on the entry margin. This also explains why the average

¹⁰As cross-validation, we show that our calculated labour income data aligns with the labour income reported by Statistics Estonia, the statistical agency that provides the official labour income statistics. More details on the sources and comparison is given in Appendix B.

¹¹This implies that the 12 permanent income groups are derived for the eight segments of workers by gender and age. Due to the conditional nature of these income groups, the income level within a single income group can be different for different segments of workers.

¹²Note that the necessary condition for entering into the analysis is that an individual has to report labour income in at least one month in the past year. We prefer average yearly income over average monthly income as it gives a better link to the existing literature using yearly data, where monthly income and transitions are not observed.

earnings growth is negative with a value of -6.0%. The job-finding and separation rates are higher for lower earnings bins, e.g. more than twice as high for the lowest income decile as for the top decile in the case of the separation rate, as shown in Table B.1. More details on the baseline statistics of our data are provided in Appendix B.

There are roughly 500 to 550 thousand individuals observed each month, and in total, there are around a million unique individuals. After balancing the panel, we have about 120 million observations. As explained in more detail in Section 3, we use Two-way clustering standard errors over individuals and dates. As this is computationally very challenging, we limit the sample to 25% of individuals randomly drawn from the full population. We ensure that the results do not depend on the specific sample. The final dataset consists of 270 thousand individuals, 299 months and 32 million observations.

External Validation: GDP Betas in Estonia and in the USA As part of the external validity evaluation, we compare our labour market data to the US. Specifically, we estimate how growth in the real labour income of workers depends on aggregate real GDP growth, i.e., the worker betas as introduced by Guvenen et al. (2017). They estimate this relationship for US data and find the workers’ exposure to economic growth to be U-shaped. The income of high and low earners is the most sensitive to aggregate growth. A similar pattern is also observed in a related paper using Swedish data by Amberg et al. (2022). However, the estimated U-shape for Sweden is not symmetric, and the lower part of the distribution is affected proportionally more.

GDP betas, β_g , are obtained by estimating the following specification for each of the earnings group:

$$\Delta y_{i,t} = \alpha_g + \beta_g \Delta GDP_t + \epsilon_{i,t}, \quad (1)$$

where $\Delta y_{i,t}$ is the mid-point average growth of total earnings, the intensive margin, entry rate or exit rate, respectively; α_g is the group-specific intercept, β_g the group-specific GDP-beta or the group-specific exposure to GDP growth. The total earnings growth from $t - 1$ to t is derived as the mid-point average growth of earnings in the balanced panel of individuals, so that the entry and exit margins are also incorporated into the total earnings growth. The intensive margin is derived as the mid-point average growth rate for those individuals who had labour income in year t and $t - 1$. The earnings groups are derived as yearly earnings at $t - 2$, ensuring non-overlapping dynamic structure, and are conditioned on age group and gender following Guvenen et al. (2017).

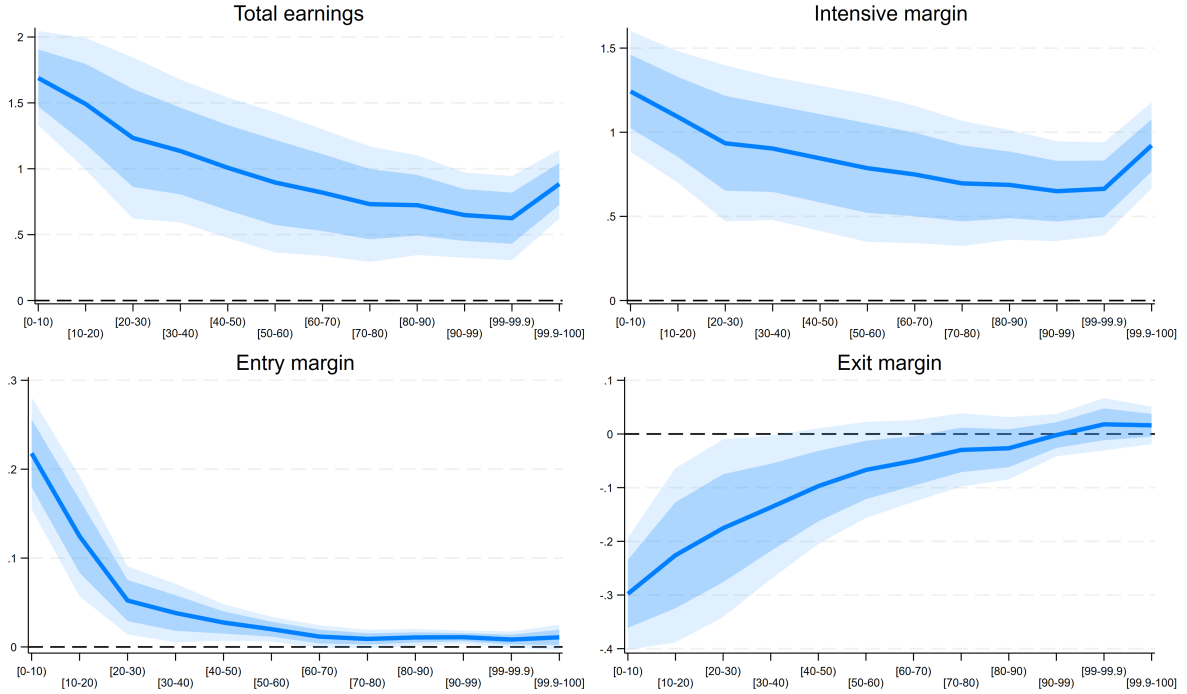


Figure 1: The GDP betas by earnings groups, yearly data, 2006-2024. GDP betas are obtained by estimating equation (1). The standard errors are clustered by individuals and dates, shaded areas represent 90 and 68% CIs.

Figure 1 shows the estimates for Estonian data, where we aggregate our monthly data to get yearly data for comparison. The figure presents the responses of total labour income, the intensive margin (labour income growth conditional on participation), the entry margin, and the exit margin. The intensive margin shown in the upper-right panel replicates the estimates of Guvenen et al. (2017) and shows very similar estimates to those found from the US data. The only deviation from the US estimates is that the upswing at the upper end is more modest, as both of the top income groups from the top 1% in the US were more sensitive to aggregate growth. This pattern is weaker in Estonian data and is only seen for the top 0.1%.¹³ Like the US data, the most affected segment in our data is at the lower end of the distribution, and this asymmetry is much more pronounced in our data. The standard errors are clustered by individuals and dates, shaded areas represent 90 and 68% CIs. In the Appendix C, we also report heteroscedasticity robust standard

¹³The level of income at these top income percentiles is also of a completely different magnitude in the US. While the top 0.1% labour income earners have a median yearly income that can be above a million of 2009 dollars, the top 0.1% labour income earners in our sample get on average 23,000 of 2015 euros per month, see Table B.1.

errors that have been used by Guvenen et al. (2017) and are, as expected, considerably smaller. Additionally, the GDP betas are also similar to that of the US in terms of gender and age groups, as discussed in Appendix C.

2.3 Monetary Policy Shocks: A High-Frequency Approach

We use a high-frequency approach to obtain the monetary policy shock series for our monthly sample from 2006M2 to 2024M12. We are focusing on the ECB’s monetary policy for our entire sample. Estonia joined the euro area in 2011. Before that, it had a fixed exchange rate as part of the European Exchange Rate Mechanism and, therefore, imported its monetary policy from the ECB during the entire sample. Another advantage of Estonia is that monetary decisions can be seen as exogenous because of the relatively small size of the country.

We compute the monetary policy surprises based on high-frequency changes in interest rates around Governing Council meetings following Jarociński and Karadi (2020).¹⁴ The changes in the interest rates are based on the Euro Area Monetary Policy Event-Study Database (Altavilla et al., 2019), which covers all Governing Council meetings between January 1999 and October 2025.¹⁵ The measure is based on the changes in the 1-month, 3-month, 6-month, and 1-year OIS rates, for which the principal component is calculated. The change in the OIS rates is the difference between the median quote before the press release and the median quote after the press conference. To focus on surprise monetary shocks and disentangle them from central bank information effects, we use the poor man’s sign restriction approach of Jarociński and Karadi (2020), which imposes restrictions on the rates and the stock market response. As a robustness check, we later also use sign restrictions based on Jarociński and Karadi (2020) and IV-based approach of Almgren et al. (2022).

Figure 2 shows the identified monetary surprises at a monthly frequency. An important point is that the monthly frequency of the Estonian administrative data enables us to use this shock series directly, without aggregating it into yearly data as is usually done in this strand of the literature. Exploiting this larger sample means we can analyse potential state dependencies. Additionally, the higher frequency allows for a cleaner composition,

¹⁴The high-frequency approach to identifying monetary policy shocks was first introduced in Kuttner (2001), Bernanke and Kuttner (2005), and Gürkaynak et al. (2005). For a recent discussion on high-frequency identification of monetary policy surprises, see, for instance, Bauer and Swanson (2023).

¹⁵We drop the Governing Council meeting on October 8, 2008, as is commonly done, since this was an unscheduled meeting enacting a coordinated interest rate cut by several central banks.

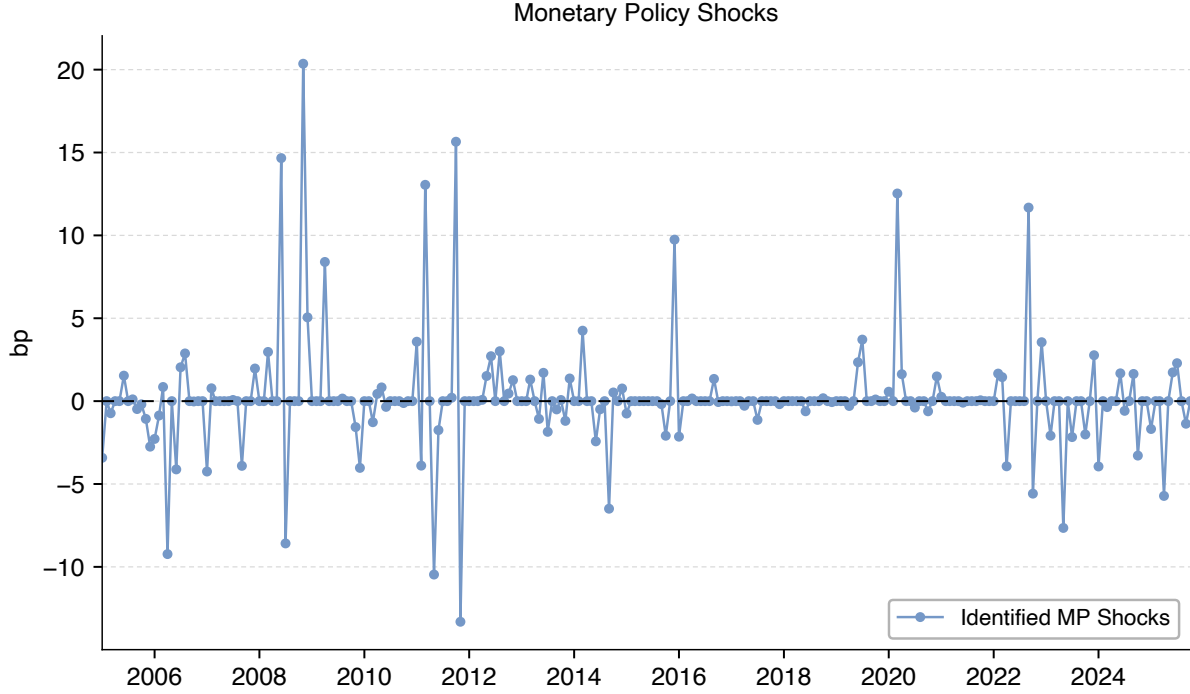


Figure 2: High-frequency identification of monetary policy shocks following Jarociński and Karadi (2020), 2005M1-2025M10.

since we can now measure the impact on the extensive and intensive margins using our monthly observations. Our data can track exactly when a person is or is not working during the year. In contrast to an analysis with yearly data, a person who works only one month and generates a positive income would count as employed throughout the year.

3 Distributional Impact of Monetary Policy

We estimate the distributional effects of monetary policy shocks using our high-frequency administrative data. The granularity of these data allows us to analyse the earnings heterogeneity channel of monetary policy and to distinguish between adjustments along the intensive and extensive margins. Specifically, we want to assess the extent to which the overall effect operates through within-employment earnings adjustments or through transitions into and out of employment. These are conceptually different adjustment mechanisms. An adjustment in earnings while remaining employed reflects the intensive margin, whereas entering or exiting employment reflects the extensive margin.

Following the related literature on the heterogeneous effects of monetary policy (e.g.,

Holm et al., 2021; Andersen et al., 2023; Amberg et al., 2022), we estimate how high-frequency monetary policy shocks affect individual wage earners across the earnings distribution. Our analysis is conducted at a monthly frequency, exploiting the high-frequency nature of our administrative records, whereas most related studies rely on annual data. An important exception is Broer et al. (2022), who use monthly German administrative labour-market histories. Relative to their data, our administrative records offer several advantages: they are not top-coded, cover all employees in both the private and public sectors, measure monthly payroll income directly rather than average earnings over employment spells, and are publicly available in near real time.

3.1 Empirical Specification

The empirical specification for estimating the impact of aggregate monetary policy shocks on individual wage-earners by labour income groups is:

$$\Delta y_{i,t+h} = \sum_{g=1}^{12} G_{i,t-2,g} \times (\alpha_g^h + \beta_g^h \Delta i_t + \gamma_g^h \Delta X_{t-1}) + M_m + \epsilon_{i,t+h}, \quad (2)$$

where the dependent variable $\Delta y_{i,t+h} = (y_{i,t+h} - y_{i,t-1}) / ((y_{i,t+h} + y_{i,t-1})/2)$ is the mid-point average growth of labour income of individual i in month $t+h$. Our approach builds on the labour economics literature, where the mid-point growth rate (Davis et al., 1998) has become a standard tool for analysing datasets such as ours.¹⁶ The advantage of the mid-point average growth rate over the conventional log difference growth rate is that it also allows zero incomes in the base or reference year to be taken into account. This means we can decompose the effect of monetary policy on earnings into the adjustment in the intensive and extensive margin by entry and exit.¹⁷ The horizon of our estimation is determined by h . Our baseline specification studies the effect over twelve months ($h = 12$), but we also evaluate alternative horizons of 6, 18, and 24 months. For the baseline estimates of $h = 12$, the effective sample of studied monetary policy shocks is 2006M2-2024M11, while the earnings data from 2005M1-2025M12 are used to obtain these estimates. The seasonality is addressed by including monthly dummies M_m , where $m = 1, \dots, 12$.

¹⁶Related applications of the mid-point average include Davis and Haltiwanger (2001) for oil shocks, Barrero et al. (2020) for the pandemic, and Decker et al. (2020), among many others.

¹⁷The mid-point average growth rate and the conventional one with only the base period value in the denominator are monotonically related. The main difference between the two growth measures is that the mid-point average growth rate is symmetric about zero and restricted to finite values -2 and 2. See more in Davis et al. (1998), pp. 26-27.

The monetary policy shock in month t is denoted by Δi_t . As positive values of the shock correspond to a contractionary shock, we expect them to lead to lower labour income growth, that is a negative coefficient for $\beta_g^h < 0$ for the different income groups, at least on average. We also allow for control variables X_{t-1} . In our baseline specification, we control for the economic activity in the domestic economy and in the Euro area using GDP growth ΔGDP_{t-1} .¹⁸ We also add domestic inflation in the month before the monetary policy shock occurs. All the growth rates of control variables are derived as year-on-year rates.

We allow heterogeneous effects of monetary policy shocks for each of our 12 permanent income groups g conditioned on age and gender following Guvenen et al. (2017). The coefficient α_g^h shows the income group-specific intercept. β_g estimates the income group-specific effect of monetary policy on labour income and allows the distributional impact of monetary policy to be assessed. The effect of control variables captured by γ_g^h can also be flexibly group-specific. The income groups $G_{i,t-2,g}$ are based on the average labour income in the last 12 months before the base earnings at y_{t-1} .¹⁹

Disentangling the total effect: The extensive and intensive margins. Equation (2) is estimated for the change in total labour income captured by $\Delta y_{i,t+h}$. To disentangle the roles of the intensive and extensive margins, we also estimate the equation separately for i) the intensive margin change where $\Delta y_{i,t+h}$ is derived only for these observations where both $y_{i,t+h}$ and $y_{i,t-1}$ have positive values; ii) the entry margin that is derived as a binary 0,1 variable taking value 1 when $\Delta y_{i,t+h}$ equals to 2 and is zero otherwise; and iii) the exit margin that is derived as a binary 0,1 variable taking value 1 when $\Delta y_{i,t+h}$ equals to -2 and is zero otherwise.²⁰ In order to incorporate non-responsive individuals with zero earnings at both t and $t+h$, the total earnings growth, entry and exit rates are amended so that these individuals are retained with zero values at t and $t+h$. The existence of some past earnings remains necessary for sample inclusion.

The effect of monetary policy on total labour income can be aggregated as the sum of the three components for the margins: the effect of the intensive margin multiplied by the fraction of workers participating in both periods $y_{i,t+h}$ and $y_{i,t-1}$; the effect of entry

¹⁸Real GDP at quarterly frequency is converted into monthly frequency using the interpolation approach of Chow and Lin (1971). The interpolation uses the related series of the unemployment rate and industrial production in line with Almgren et al. (2022).

¹⁹This implies that a first-time worker does not affect the entry margin. The first-year labour income of a newly entering worker is used to derive the labour income group.

²⁰There can be periods as long as one month of no labour income in the data because of vacations. We do not want to treat these as entries or exits. To avoid this, we impose that the spell of the employment or non-employment status needs to last for at least one month.

multiplied by two; and the effect of exit multiplied by minus two.

Two-way clustered standard errors. We report two-way clustered standard errors by individual and time for statistical inference. This approach addresses autocorrelation and heteroscedasticity flexibly in both dimensions, by individuals and time, and is a conservative estimate providing wide confidence bounds. The majority of the statistical uncertainty around the estimates originates from clustering by time and not from clustering by individuals. This time-clustering of standard errors is introduced because our treatment variable, monetary policy shock, does not vary for individuals within a calendar month.

One issue is that two-way clustering is computationally demanding. This poses a challenge, as the total dataset has roughly 500 to 550 thousand individuals observed each month, and in total, there are about a million unique individuals over the entire sample. After balancing the panel, we have about 120 million observations. In order to meet the computational capacity, we rely on a random draw of 25% of the sample for the first ten earnings bins and retain the remaining two upper bins that account for the top 1% of earners. For this reason, we also introduce sampling weights to adjust for oversampling of the top 1%. The final dataset consists of 270 thousand individuals, 299 months and 32 million observations. We ensure the robustness of our findings by taking alternative draws and validating our findings.

3.2 Results

The effect of a contractionary monetary policy shock on earnings is shown in Figure 3, which is associated with a 1.1% decline in accumulated total earnings after 12 months across all the earnings bins.²¹ The shock has a negative effect on earnings throughout the entire earnings distribution. However, the effect is unequally distributed, as low-wage workers are much more strongly affected than high-wage workers. Note that we also observe a slightly stronger exposure for the top 0.1%. However, the difference compared to the top 1% and 10% is very small. Taken together, our results show that workers' exposure to a monetary policy shock is heterogeneous, with low-income workers most affected.

Disentangling the effect shows that workers are unequally affected by the different adjustment margins, see also Figure 4. Low-wage workers are affected most by the extensive

²¹This measure also reflects adjustments along the entry and exit margins. The effect operating through the intensive margin alone amounts to a 0.3% decline in earnings. These results can be interpreted in the context of participating workers, for whom the average wage growth is 3.8% over the same horizon.

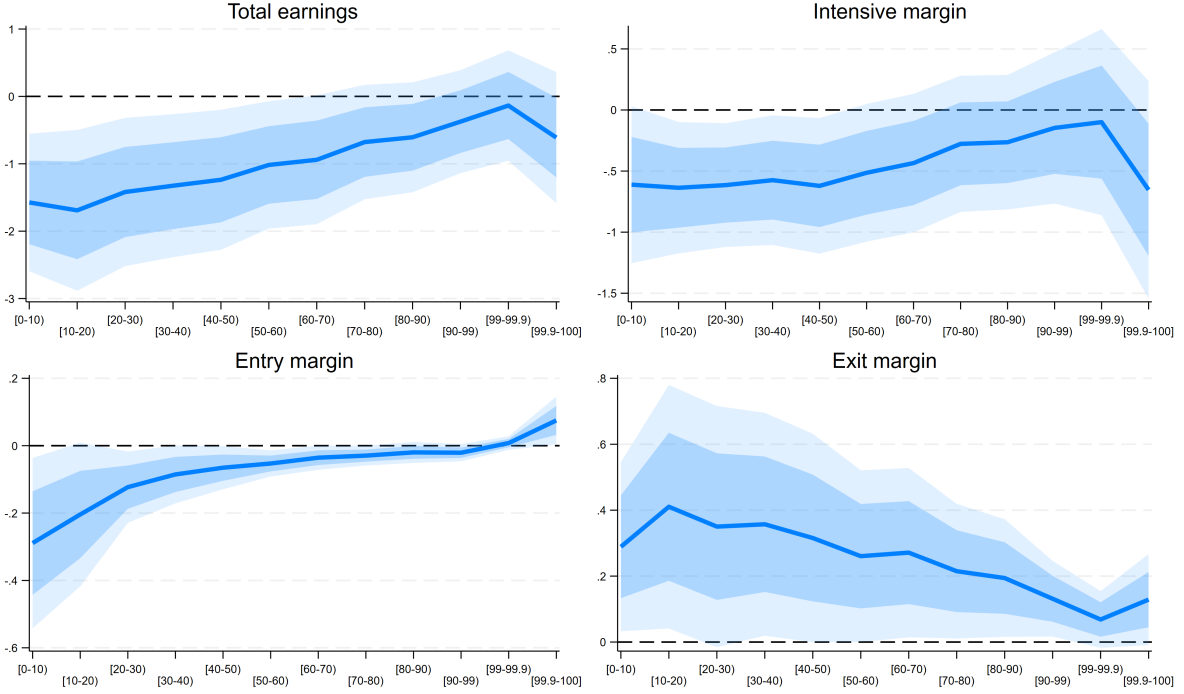


Figure 3: The effect of a monetary policy shock (1SD) on earnings growth (%) by earnings groups, 2006M2-2024M21. The effect on total earnings, the intensive margin, the entry margin and the exit margin are shown. Each value in a figure represents estimates of $(\beta_g^h * 0.032) \times 100$ for horizon $h = 12$ in equation (2), indicating the effect of 1SD monetary policy shock on the accumulated earnings growth of group g by month 12. The standard errors are clustered by individuals and dates, shaded areas represent 90 and 68% CIs.

margin, as shown by their strong response through the entry and exit margins; middle-group workers from the fourth decile to the ninth decile are affected by both the intensive and extensive margins, with the extensive margin becoming less important as earnings rise; and the highest earnings group, the 0.1%, are again equally affected by the extensive and intensive margin. Low-wage workers are consequently most affected by monetary policy because they transition in and out of employment. In contrast, high-wage workers hardly enter or exit employment due to monetary policy, but experience real wage adjustments.

An important observation is that the adjustment patterns differ substantially between monthly and yearly data, as highlighted by the decomposition results in Figure 4. The yearly data misses substantial within-year transitions into and out of employment.²² A

²²Monthly frequency is particularly useful because employment transitions are naturally captured at that horizon. By contrast, at the annual frequency, a worker who exits employment for several months

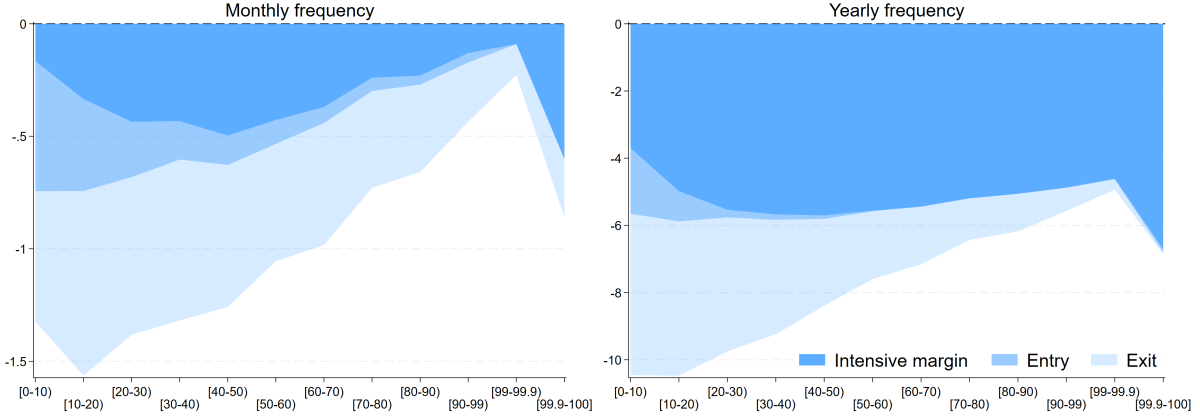


Figure 4: Comparison of the decomposition in the different margins between monthly data (left panel) and annual data (right panel), 2006-2024. Each value in a figure represents the effect of a 1SD monetary policy shock by margin and group g , accumulated for horizon $h = 12$. 1SD corresponds to a 3.2 basis point shock at monthly frequency and a 9.7 basis point shock at yearly frequency.

majority of the monetary policy effect on low-wage people originates from the intensive margin by construction.²³ Specifically, we find that the intensive margin accounts for one-third of the total effect in monthly data, but for two-thirds of it in yearly data. Using our higher frequency data thus identifies the impact of the intensive and extensive margins better. In contrast, annual data can blur the distinction between the two margins and overstate the role of the intensive margin. More details, including the total earnings response as well as the intensive and extensive response to a monetary policy shock for a yearly horizon, are in Appendix D.1.

Our baseline specification studies how monetary policy affects labour income growth over twelve months. Additionally, we also estimate equation (2) with alternative horizons. We use labour income growth rates over 6, 18 or 24 months ($h = 6, 18, 24$). The results are in line with related literature on aggregate dynamics. Most of the effect of monetary policy materialises by month 12, and it peaks in month 18 or 24, as can be seen in Figure 5. The pattern along the earnings distribution is also stable. Finally, we also show that the effects depend on gender and age group in Appendix D.2. Men and younger workers are more exposed to monetary policy shocks than women and older workers.

and later re-enters within the same year may still appear as having experienced only an earnings change, even though part of the adjustment operated through becoming unemployed.

²³Our estimates with yearly data for the intensive margin are close to the estimates of Amberg et al. (2022).

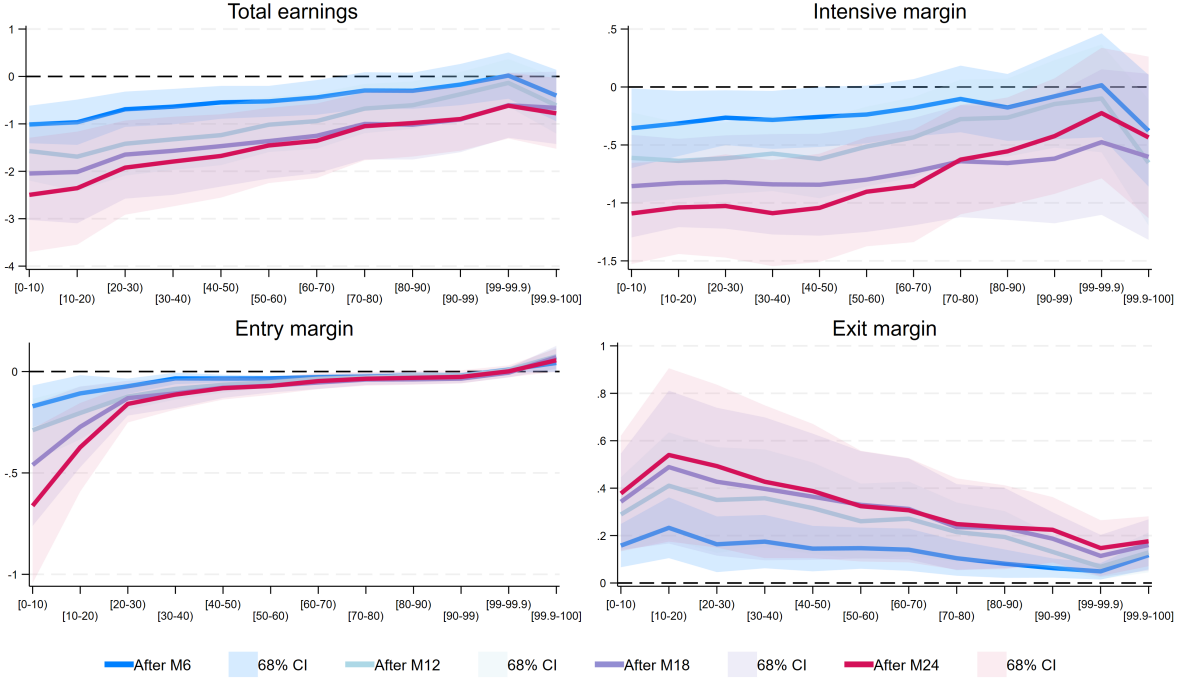


Figure 5: The effect of monetary policy shock on earnings by earnings groups, local projections by month 6, 12, 18, 24, 2006M2-2024M12. Please refer to Figure 3 for notes.

3.3 Robustness Checks

We conduct a range of robustness checks using alternative monetary policy shocks (including an IV approach), different sets of control variables, nominal wages instead of real wages, alternative confidence bounds, alternative growth measures instead of mid-point average growth, and a different random draw from the full sample. Figure 6 summarises most of these results.

Alternative Monetary Policy Shocks. We consider alternative measures of the monetary policy shock by adding and relaxing the restrictions for identifying shocks.²⁴ We first add restrictions by considering the sign-restriction approach of Jarociński and Karadi (2020).²⁵ We also employ the instrumental variable technique where policy rate EONIA

²⁴Brennan et al. (2024) discusses how different data series and methods create differences in the series of identified shocks.

²⁵Like with the poor man's sign restriction for our main series, the goal of this is to focus on surprise monetary shocks and disentangling them from central bank information effects. In this case, we use their sign restriction approach, which imposes restrictions on the rates and the stock market response. The

is instrumented with changes in the 3-month OIS rate over the monetary policy event window. This approach follows Almgren et al. 2022 and allows for interpreting the results in terms of basis point changes in the policy rates, see more on methodology in D.3. The result of the distributional effects of monetary policy by alternative measures of monetary policy is shown in the upper panel of Figure 6. The estimated response of earnings income is very close to our baseline estimates and also shows that the monetary policy has distributional effects. Compared to the baseline, the instrumental variable estimates suggest a stronger U-shape for the top 0.1%. The average effects are also size-wise very close; the average effect of a one standard deviation increase in EONIA over all the earnings bins is -1.4%, while our baseline estimates indicate a 1.1% decline in earnings due to a contractionary shock of one standard deviation. However, we also note that the interpretation of both approaches is not identical. While both specifications use high-frequency monetary policy surprise, the baseline specification estimates the effect of the surprise itself, whereas the IV specification estimates the effect of the policy-rate change induced by a surprise.

Alternative Controls. We also add the euro area geopolitical risk measure (Bondarenko et al., 2025) as an additional control variable to the baseline specification.²⁶ The advantage of this control is that geopolitical risk is unlikely to be affected by monetary policy shocks, and so it can enter contemporaneously with a monetary policy shock. Controlling for it can be very important because geopolitical risk has affected the dynamics of income in Estonia substantially. This was especially so when inflation was high after the start of the full-scale war in Ukraine in 2022, and also, to some extent, stemmed from the associated surge in energy prices. The results are shown in the left panel in the second row of Figure 6. The results are very assuring because they are very close to our baseline estimates.

We also incorporate an additional lag for the macro controls in the estimation. The results are consistent, as can be seen in the right panel in the second row of Figure 6.

Nominal wages instead of real wages We re-evaluate the effect of monetary policy using nominal wages instead of real wages. The results with nominal wages as the dependent variable are shown in the third row, left panel of Figure 6. Whether the dependent variable is deflated or not has a limited impact on the results. In fact, the response with

median response then provides the shock series.

²⁶The measure builds on the methodology of Caldara and Iacoviello (2022), but tailors it to the euro area by using local news sources, following Bondarenko et al. (2024).

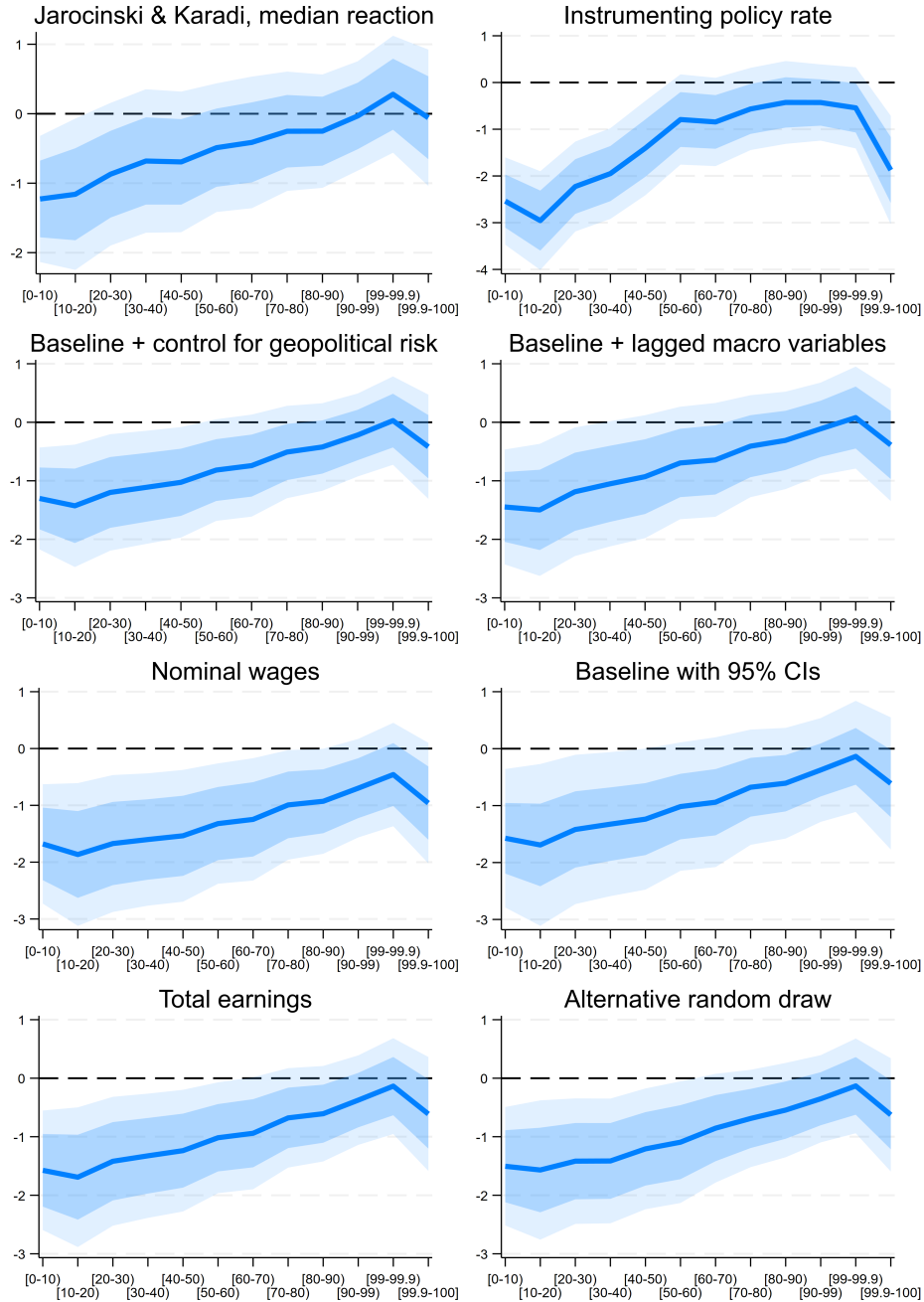


Figure 6: The effect of a monetary policy shock (1SD) on earnings growth (%) by earnings groups, alternative estimates, 2006M2-2024M12. The effect of 1SD monetary policy shock on the accumulated earnings growth of group g by month 12 are shown. 1SD shock equals to 1.6bps for IV, 3.3bps for median reaction and 3.2bps for the rest of the figures. Please see Figure 3 for additional notes.

nominal wages shows there to be a slightly stronger effect across the entire distribution. This comes from how inflation affects real wages. For instance, the negative effect on nominal wages is compensated for by lower inflation.

Confidence intervals We also report the 95% confidence interval. While the bounds are larger, the stronger exposure of low-income households is robust, as can be seen in the right panel in the third row of Figure 6.

Mid-point growth rate As a robustness test, we estimate equation (2) using the past earnings in the denominator for the total earnings response and with the log-change based growth rate for the intensive margin. For total earnings, the results are close to a measure that uses lagged earnings in the denominator, as the left lower panel of Figure 6 highlights.²⁷ However, this alternative measure is more sensitive to outlier trimming: without trimming, the confidence intervals widen substantially for the lowest decile, while stricter trimming narrows them. When we only want to assess the intensive margin (employed workers), the mid-point growth rate is quantitatively very close to the log-change measure, as shown in Figure D.4 in the appendix.

Alternative random draw As discussed above, we work with a random sample because computing two-way clustered standard errors on the full dataset of 120 million observations is computationally infeasible for us. Importantly, our results are not sensitive to the particular random draw. For example, the right lower panel of Figure 6 reports the overall effect based on an alternative random sample.

4 Aggregate Implications of the Earnings Heterogeneity Channel

This section evaluates the aggregate implications of the earnings heterogeneity channel. In particular, we determine how different exposures to monetary policy across the earnings distribution amplify or dampen the response of aggregate consumption. To achieve this, we must complement our individual-level estimates of earnings risk to monetary policy shocks with information on the MPC. We again exploit the advantages of using our Estonian

²⁷For the specification using lagged annual earnings in the denominator, we trim observations with 1% of the lowest and 1% of the highest earnings growth.

data by merging our administrative data with household-level MPC estimates from the Household Finance and Consumption Survey (HFCS) for Estonia.²⁸ Using this merged dataset, we evaluate the aggregate MPC in response to a monetary policy shock. We also analyse the impact of a monetary policy shock on total earnings and the Gini coefficient.

4.1 The Aggregate Marginal Propensity to Consume

The aggregate MPC for a monetary policy shock can be disentangled into components. These are the income-weighted average MPC and the income-weighted covariance between the relative household-level exposure to a monetary policy shock and the MPC:

$$\text{Aggregate MPC}|_{mp} = \left(\sum_j \frac{dC_j}{dE_j} \frac{dE_j}{dmp} \right) \frac{dmp}{dY} = \left[\left(\sum_j \frac{E_j}{Y} \frac{dC_j}{dE_j} \right) + \text{cov} \left(\frac{dC_j}{dE_j}, \gamma_j^{mp} \right) \right], \quad (3)$$

where E_j is the labour income of household j , C_j is the consumption of household j , Y is the aggregate labour income, and $\gamma_j^{mp} = \frac{Y}{E_j} \frac{dE_j}{dmp} \frac{dmp}{dY}$ is the relative earnings exposure to a monetary policy shock. We focus here only on the earnings channel. The derivation is based on a simple model of Patterson (2023), which we extended to directly account for monetary policy shocks. The consumption function of household j depends on the monetary policy shock, that is $C_j = C_j(E_j(Y(mp)), \theta_j)$. We focus here entirely on the earnings channel of monetary policy and abstract from other channels. We assume that there is no capital or profits so that we have $Y = \sum_j E_j$. The detailed derivation is delegated to Appendix E of the paper.

The first term on the right-hand side is the income-weighted average MPC of households. The second term captures the income-weighted covariance between household-level exposure to a monetary policy shock and household-level MPCs.²⁹ The second term determines the contribution of the earnings heterogeneity channel to the monetary policy shock transmission. A positive covariance between the household-level MPC and relative earnings exposure, $\text{cov}(\cdot) > 0$, amplifies the monetary shock due to earnings heterogeneity. A negative correlation, in contrast, $\text{cov}(\cdot) < 0$, would dampen the impact. Similarly, if the household-level response to a monetary policy shock is unrelated to their MPC, $\text{cov}(\cdot) = 0$, the earnings heterogeneity channel would not contribute to the transmission of monetary

²⁸The Household Finance and Consumption Survey is an ECB-coordinated survey of household assets, liabilities, and consumption. It is analogous to the US Survey of Consumer Finances.

²⁹Note that we express this equation at the household level due to the availability of MPC estimates. When describing the data below, we explain how we adjust our initial estimates to have them on a household level.

policy.

4.2 Microsimulation Approach and Merged Dataset with MPCs

To determine the aggregate MPC, we use a microsimulation approach. In particular, we construct the aggregate MPC for monetary policy shocks using the following expression:

$$\text{Aggregate MPC}|_{mp} = \left(\sum_j \frac{dC_j}{dE_j} \frac{dE_j}{dmp} \right) \frac{dmp}{dY}. \quad (4)$$

An important advantage of our approach is that, given our detailed dataset, we can directly account for heterogeneity in individuals' exposure to monetary policy shocks. This contrasts with some related work, in which the microsimulation response is derived under the assumption that all households face the same exposure.

However, we do not have individual- or household-level estimates of the marginal propensity to consume in our administrative data. To obtain MPC estimates, we therefore use the HFCS for Estonia conducted in 2021 and 2024. These surveys provide a household-level MPC estimate based on the unexpected windfall gain question.³⁰ It has been shown that such self-reported MPC measures provide similar MPC estimates to those revealed from actual behaviour and are informative in predicting actual spending (Parker and Souleles 2019). Table 1 shows the average MPC for the different earnings percentiles, averaging the estimates over the two waves of the survey.³¹ We observe a substantial heterogeneity in the MPCs over the income distribution, like in related literature by Jappelli and Pistaferri (2014).³²

We link the sample of Estonian households that were interviewed in the HFCS with our administrative data-based estimates of responsiveness to monetary policy at the individual level.³³ We exploit that the 2021 and 2024 wave of the HFCS reports the income from

³⁰The exact phrasing of the MPC question is (HFCS question HIZ040a): "Imagine you unexpectedly receive money from a lottery, equal to the amount of income your household receives in a month. What per cent would you spend over the next 12 months on goods and services, as opposed to any amount you would save for later or use to repay loans" The reference unit is the household.

³¹Note that the results from two survey waves are very similar, see Figure F.2, and our conclusions do not depend on whether we use the average across survey waves or any particular survey wave.

³²Our average MPC estimate is 37% for all the households with at least one member of prime working age in employment. This is somewhat lower than the estimates using exactly the same wording of the windfall gain question in Jappelli and Pistaferri 2014 and the revealed-preference estimates from spending after income shocks such as unemployment or tax rebates in Patterson 2023.

³³Note that we merge the two highest earnings groups as the HFCS survey data does not allow us to observe the earnings group of the top 0.1%.

the last calendar year. Note that we include only those households that have at least one member of prime working age, 26-65, who has received labour income. Our final sample contains 1527 households in 2021 and 1716 in 2024. Thus, the HFCS survey provides us directly with the household-specific MPC estimates:

$$\frac{dC_j}{dE_j}. \quad (5)$$

The next step in our microsimulation is to derive dE_j/dmp . To do so, we use the estimated heterogeneous earnings responses to monetary policy shocks at the individual level, denoted by dE_i/dmp . Specifically, we use the estimated exposure coefficients, β_g^h , by earnings group g and horizon h , separately across socio-economic groups. In total, we have estimates for this heterogeneous exposure, β_g^h , for 11 earnings groups within eight socio-economic groups (gender and four age groups), which yields a total of 88 coefficients. Given a monetary policy shock of size mp , we then translate the estimated response into an implied earnings change at the individual level. Using the definition of the mid-point growth rate, the implied earnings change for individual i is given by

$$\frac{dE_i}{dmp} = (((2 + \beta_g^h \times mp)/(2 - \beta_g^h \times mp)) - 1) * E_i, \quad (6)$$

where E_i denotes the individual's baseline earnings and mp the size of the monetary policy shock.³⁴ In the simulation, we calibrate this to a one-standard-deviation contractionary monetary policy shock, i.e. $mp = 0.032$. We then aggregate the resulting individual-level earnings changes to the household level to obtain the desired component:

$$dE_j/dmp = \sum_i \frac{dE_i}{dmp}. \quad (7)$$

We then combine these estimates for all observed households in the HFCS dataset and compute

$$\sum_j \frac{dC_j}{dE_j} dE_j/dmp \quad (8)$$

The final step is to adjust this expression by $\frac{dmp}{dY}$. Specifically, this term can be written

³⁴Note that this involves a slight abuse of notation, as we use mp to denote the shock directly inside the equation, in line with our baseline specification.

as

$$\frac{dmp}{dY} = \left(\sum_j \frac{dE_j}{dmp} \right)^{-1} \quad (9)$$

where we use $Y = \sum_j E_j$. Note that we have already derived dE_j/dmp in equation (7). We have now all components to conduct our microsimulation approach.

Note that our aggregate MPC should be interpreted at an annual horizon rather than as a month-by-month consumption response as the MPC question in the HFCS focuses on the next 12 months. Specifically, we take the cumulative earnings effect over the first 12 months after the monetary policy shock and map it into a 12-month consumption effect using the annual MPC reported in the HFCS. This provides an estimate of the aggregate consumption implications of the earnings heterogeneity channel, but it does not pin down how spending adjustments are distributed within the year.³⁵

4.3 Aggregate Results

We calculate and analyse the aggregate MPC estimates for a monetary policy shock, as shown in Table 1. We first characterise a baseline scenario in which we assume that monetary policy has an equal effect on all earnings groups. This corresponds to the earnings-weighted MPC captured by the first term on the right-hand side of equation (12) and abstracts from the earnings heterogeneity channel. The income-weighted aggregate MPC equals 0.324, as shown in column (1). In such a world, a contractionary monetary policy shock of one standard deviation would lead to a decline in consumption of 0.35%. This measure is calculated as the response of total earnings to a monetary policy shock, which is 0.011 (see also Table D.1) times the income-weighted MPC (more precisely, $0.0107 \cdot 0.324$).

However, the earnings groups are not affected equally by the monetary policy shocks, as shown in the previous sections. Low-wage groups with high MPCs are affected much more by the monetary policy shock. There is a positive correlation between the earnings elasticity with respect to monetary policy and the MPC, as shown in detail in Figure F.2 in the Appendix. Since households are not equally affected by monetary policy, the aggregate MPC rises to 0.350. This aggregate MPC calculation assigns around 7.4% of the impact

³⁵While the HFCS elicits spending over a 12-month horizon and therefore does not identify the exact within-year timing, evidence from the literature (e.g. Fagereng et al., 2021) suggests that consumption responses to income shocks are typically front-loaded.

Table 1: The MPCs and monetary policy effects by household earnings groups

Earnings percentile	MPC	(1) Earnings weight	(2) Relative response to MP	(3) Earnings $\times$ MP weight
[0 – 10)	0.451	0.014	1.996	0.027
[10 – 20)	0.452	0.032	1.832	0.059
[20 – 30)	0.445	0.047	1.521	0.071
[30 – 40)	0.374	0.060	1.381	0.083
[40 – 50)	0.372	0.072	1.277	0.093
[50 – 60)	0.331	0.087	1.126	0.097
[60 – 70)	0.334	0.106	1.161	0.122
[70 – 80)	0.349	0.130	1.065	0.138
[80 – 90)	0.311	0.169	0.854	0.144
[90 – 99)	0.270	0.230	0.643	0.148
[99 – 100]	0.171	0.054	0.322	0.017
Weighted Aggregate MPC		0.324	-	0.350
Consumption response to MP shock		-0.35%	-	-0.38%
Earnings Gini response to MP shock		0.00%	-	0.35%

Notes: The table shows the MPC, the income weight (1), the relative response to monetary policy (2) and the weight adjusted by the relative response to monetary policy shocks (3). The last column is obtained by multiplying columns (1) and (2). The aggregate MPC is calculated using the different weighting schemes numbered (1) or (3), which come from earnings or earnings adjusted with the heterogeneous exposure to the monetary policy shock. The last two rows show the response of consumption and inequality to a contractionary monetary policy shock of one standard deviation (3.2 bps). The response is conditioned on the different weighting schemes and the size of the monetary policy shock. Individuals' response to the monetary policy shocks from 2006M2-2024M12 and households' MPCs from 2021 and 2024.

of monetary policy on aggregate consumption to the earnings heterogeneity channel. A contractionary monetary policy of one standard deviation shock leads to a decline in consumption 0.38% ($0.0107 \cdot 0.350$). Furthermore, a monetary policy shock of one standard deviation increases inequality by 0.35%, which we measure using the Gini coefficient for earnings³⁶.

4.4 Intensive and Extensive Margins

We can also distinguish between the intensive and extensive margins. For this, we condition the aggregate MPC for monetary policy shocks on the earnings channel operating

³⁶The earnings Gini was 0.4029 and 0.4043 in our sample in 2020 and 2023, respectively.

through either the intensive or the extensive margin:

$$\text{Aggregate MPC}|_{mp} = \left(\sum_j \frac{dC_j}{dE_j^k} \frac{dE_j^k}{dmp} \right) \frac{dmp}{dY} = \left[\left(\sum_j \frac{E_j}{Y} \frac{dC_j}{dE_j^k} \right) + \text{cov} \left(\frac{dC_j}{dE_j^k}, \gamma_j^{mp^k} \right) \right], \quad (10)$$

where the superscript $k \in \{ext, int\}$ indicates whether the earnings change stems from the extensive or the intensive margin. This requires to adapt the microsimulation accordingly. Specifically, we use the estimates of β_g^h in (6), which we computed separately for the extensive and intensive margins.

We find that the aggregate MPC for monetary policy shocks is 0.343 along the intensive margin, implying that 5.5% of the overall monetary policy effect is accounted for by the earnings heterogeneity channel in that case. When we instead focus on the extensive margin, the aggregate MPC is slightly higher, at 0.353, which implies that somewhat more than 8% of the monetary policy effect operates through the earnings heterogeneity channel. Overall, both the intensive and the extensive margin amplify the response to monetary policy shocks, but the amplification is stronger through the extensive margin. For details, see F.1 in the appendix.

5 Inflation Surges and the Distributional Impact of Monetary Policy

One crucial advantage of our near-real-time high-frequency data, especially relative to annual or only infrequently updated data, is that they can study relatively short-lived but influential periods. In particular, state dependence and nonlinearities may shape the transmission of monetary policy during some periods. For example, Gargiulo et al. (2024) show, using aggregate US data, that the effects of monetary policy depend on the level of inflation. Specifically, they find that monetary policy has more persistent and longer-lasting effects on unemployment in a high-inflation environment.

Applying our data to the recent inflation surge, we use our data to analyse how the inflation environment affects the earnings heterogeneity channel of monetary policy. Importantly, we can study whether the level of inflation shapes the distributional effects of monetary policy, thereby going beyond state-dependent results at the aggregate level.

5.1 Empirical Specification with Inflation Regimes

We now extend our empirical specification to account for the impact of the inflation environment. We define a high-inflation regime as a period in which inflation exceeds 7%. This threshold corresponds to inflation being half a standard deviation above its mean.³⁷ To evaluate how the earnings heterogeneity channel of monetary policy depends on the inflation environment, we estimate the distributional impact in a low-inflation and high-inflation regime:

$$\begin{aligned} \Delta y_{i,t+h} = & (\Pi_{t-1} < 7\%) \times \sum_{g=1}^{12} G_{i,t,g} \times (\alpha_g^{h,L} + \beta_g^{h,L} \Delta i_t + \gamma_g^{h,L} \Delta X_{t-1}) \\ & + (\Pi_{t-1} \geq 7\%) \times \sum_{g=1}^{12} G_{i,t,g} \times (\alpha_g^{h,H} + \beta_g^{h,H} \Delta i_t + \gamma_g^{h,H} \Delta X_{t-1}) + M_m + \epsilon_{i,t+h}. \end{aligned} \quad (11)$$

We now have two coefficients for the monetary policy impact: $\beta_g^{h,L}$ captures the impact of monetary policy for income group g in the low-inflation regime (inflation below 7%), and $\beta_g^{h,H}$ captures the impact in the high-inflation regime (7% or above).³⁸ We define the inflation regime using the annualised inflation rate with respect to $t - 1$ because our objective is to estimate the effect of a monetary policy shock conditional on the inflation environment prevailing when the shock hits. This approach is consistent with the logic of the state-dependent local-projection literature (e.g. Tenreyro and Thwaites, 2016; Alpanda et al., 2021; Cloyne et al., 2023) and with recent econometric work emphasizing that the interpretation of state-dependent responses depends on how the state variable is determined (Gonçalves et al., 2024). It is also in line with the recent inflation-regime work of Gargiulo et al. (2024). The macro controls are also interacted with the inflation regime to account for the potentially heterogeneous impact of aggregate variables on earnings at different levels of inflation. The regime-specific intercepts account for the time-invariant growth in a regime.

The parameters $\beta_g^{h,L}$ and $\beta_g^{h,H}$ determine the role that the inflation regime might play in the earnings heterogeneity channel of monetary policy. While we expect the direction of the

³⁷While the 7% threshold may seem large, inflation in Estonia has been, on average, well above 2%, as can be seen in Figure A.1.

³⁸One concern in our analysis could be that there is a systematic relationship between the monetary shocks and the inflation regime. But, comparing the data shows that there does not seem to be a relationship between the direction or size of the high-frequency monetary policy shock and the inflation regime.

shock to be rather constant across the regimes, so that $\beta_g^{h,L}$ and $\beta_g^{h,H}$ are mostly similar, the magnitude of the impact may be very different. For instance, if $|\beta_g^{h,H}| > |\beta_g^{h,L}|$, monetary policy has a stronger effect for income group g in the high-inflation regime than in the low-inflation regime. The shape of the distribution may also be affected. We also disentangle the effects to evaluate the intensive and extensive margins separately, allowing response mechanisms to be different in a low- and high-inflation environment.

5.2 Results by Inflation Regime

Figure 7 presents the distributional effects of a monetary policy shock in a low- and high-inflation regimes. Given the short-lived nature of the high-inflation regime, we report the 68% confidence intervals for the two-way clustered standard errors for each regime. One main finding is that the impact of monetary policy on earnings is amplified in the high-inflation period. The response of total earnings is considerably more negative across almost the entire earnings distribution. In particular, low-wage workers are affected substantially more strongly than middle- or high-wage workers in a high-inflation environment.³⁹ In addition, the exposure displays a mild U-shape, as the top 0.1% of earners are also affected to some extent more strongly in the high-inflation regime. By contrast, in the low-inflation regime, the effect is quantitatively smaller across almost the entire earnings distribution, and exposure declines roughly monotonically with earnings. Thus, with the exception of the very top of the distribution, especially the top 0.1% of earners, the effects are stronger in the high-inflation regime.

The inflation regime also impacts the adjustment margins. The lowest earnings deciles are more exposed to the monetary policy shock in the high-inflation regime primarily because of the extensive margin. Entries and exits in the labour market dominate the response. For instance, for the lowest decile, 87% of the response is explained by the entry and exit margins, as Table D.2 in the appendix details. In the low-inflation regime, though, the exposure of the low-wage workers stems to a much larger extent from the intensive margin. On average, the intensive margin accounts for slightly more than half of the response in the low-inflation regime and slightly more than a fifth in the high-inflation regime. Figure 8 summarises the overall effect and the decomposition in the different margins between periods of low and high inflation.

³⁹Another relevant mechanism is that inflation rates differ across poor and rich households, so that poorer households may have been more strongly exposed during the recent inflation surge (e.g., Hochmuth et al., 2023).

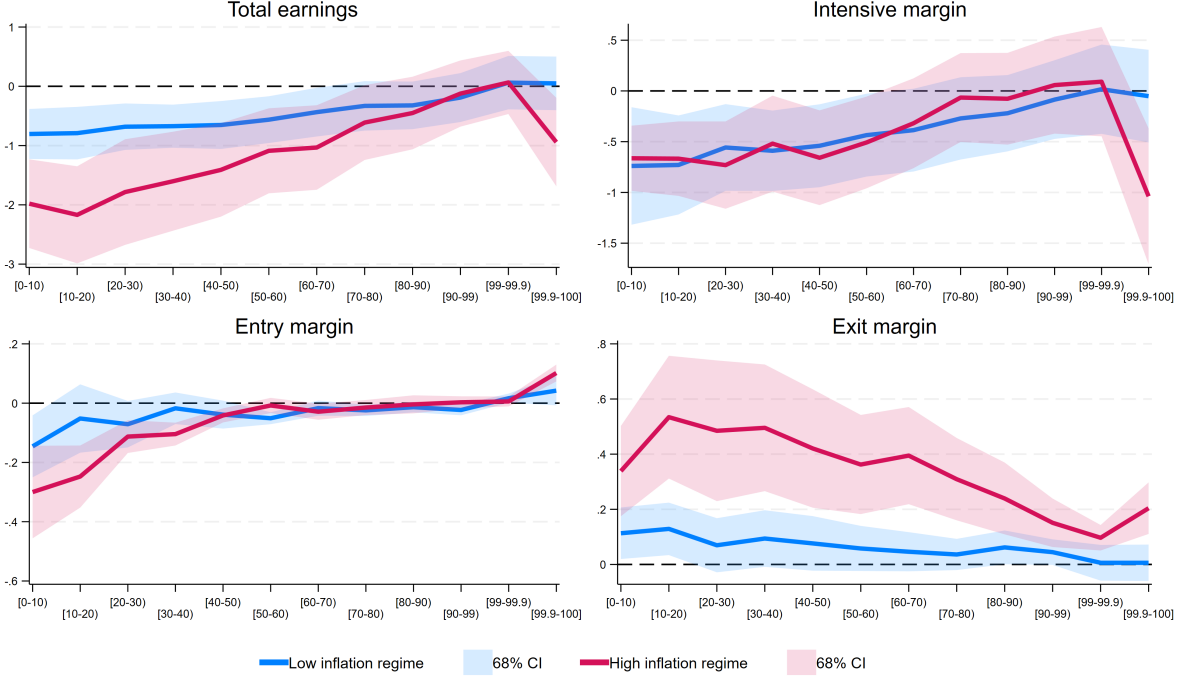


Figure 7: The effect of monetary policy shock (1SD) on earnings growth (%) by inflation regime and earnings groups, 2006M2-2024M12. Estimates of equation (11). Please refer to Figure 3 for more notes.

At the same time, we want to emphasise that we observe only two periods of very elevated inflation in our sample, namely a period around the financial crisis and a period following the pandemic. Since both episodes were characterised by substantial macroeconomic turbulence, the results should be interpreted with appropriate caution.

5.3 Aggregate Implications by Inflation Regime

We can also condition the aggregate impact on the inflation environment.

$$\text{Aggregate MPC}^s|_{mp} = \left(\sum_j \frac{dC_j^s}{dE_j^s} \frac{dE_j^s}{dmp} \right) \frac{dmp}{dY^s} = \left[\left(\sum_j \frac{E_j^s}{Y^s} \frac{dC_j^s}{dE_j^s} \right) + cov \left(\frac{dC_j^s}{dE_j^s}, \gamma_j^{mp^s} \right) \right], \quad (12)$$

where $\gamma_j^{mp^s} = \frac{Y^s}{E_j^s} \frac{dE_j^s}{dmp} \frac{dmp}{dY^s}$. The superscript s indicates whether the economy is in a low inflation period, e.g., $s = L$ or in a high inflation period, $s = H$.

In particular, both the earnings response to a monetary policy shock and the household-

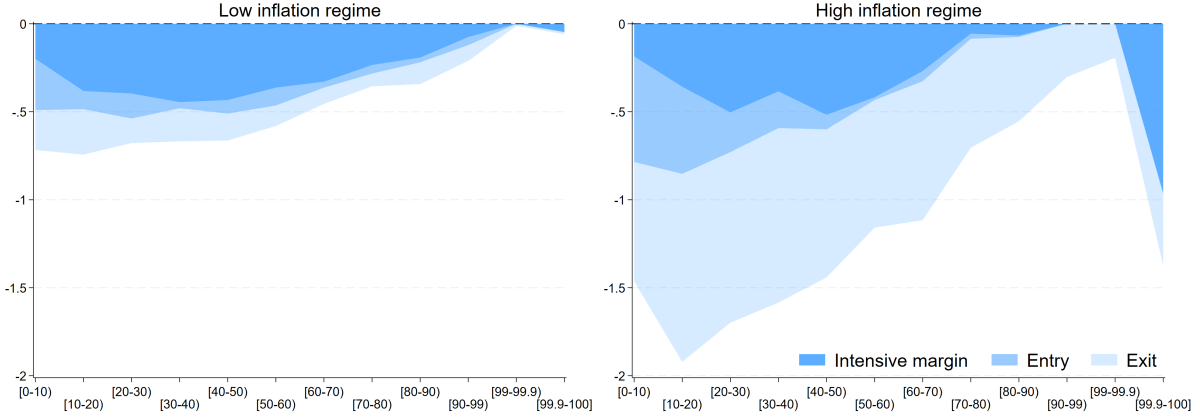


Figure 8: Comparison of the decomposition in the different margins between low inflation (left panel) and high inflation (right panel) regimes, 2006M2-2024M12.

level marginal propensity to consume may differ across inflation regimes. Using the inflation-dependent estimates, the aggregate MPC in equation (12) is calculated with the elasticity γ_j^L and γ_j^H for the low and high inflation regimes, respectively. The survey data on MPC were collected either shortly before the inflation surge, between January and August 2021, or shortly after the inflation surge, between February and June 2024. There is evidence that MPCs are quite stable over the business cycle (Patterson 2023), and this is confirmed by our data. While we would expect the MPCs to increase in a high-inflation environment, especially among the low-income groups as they spend the largest part of their income on essentials, the MPC distribution over earnings groups changes only marginally between the two waves, see Figure F.2. We take the household-level MPCs before the inflation surge as a proxy for MPCs in the low inflation environment and MPCs after the inflation surge as a proxy for MPCs in the high inflation environment. However, we believe this is a cautious assumption, as inflation declined to below 7% threshold in July 2023 and households could have adjusted their high-inflation MPCs already by 2024. Therefore, we may downward bias the aggregate MPC for the high-inflation environment, resulting in a lower estimate. It should still provide a useful starting point because of the regime-varying monetary policy exposure and MPCs.

The results are shown in Table 2. The covariation between the exposure to a monetary policy shock and the MPC is stronger during the high-inflation regime than during the low-inflation regime, although the differences are small. As a consequence, the aggregate MPC is larger in the high-inflation regime. The stronger response of the aggregate MPC is due to two reasons. The first is that the low-wage groups are relatively more affected

Table 2: The MPCs and monetary policy effects by household earnings groups

Earnings percentile	Low inflation			High inflation		
	(1) Earnings weight	(2) Relative response	(3) MP weight	(4) Earnings weight	(5) Relative response	(6) MP weight
[0 – 10)	0.013	1.843	0.025	0.014	2.474	0.034
[10 – 20)	0.033	1.770	0.058	0.031	2.308	0.072
[20 – 30)	0.046	1.484	0.069	0.047	1.827	0.085
[30 – 40)	0.061	1.339	0.081	0.060	1.552	0.093
[40 – 50)	0.073	1.303	0.095	0.072	1.379	0.100
[50 – 60)	0.086	1.101	0.095	0.087	1.183	0.103
[60 – 70)	0.105	1.117	0.117	0.106	1.207	0.128
[70 – 80)	0.130	1.120	0.146	0.130	1.025	0.133
[80 – 90)	0.169	0.879	0.148	0.169	0.744	0.126
[90 – 99)	0.229	0.648	0.148	0.232	0.508	0.118
[99 – 100]	0.056	0.331	0.018	0.053	0.175	0.009
Weighted Aggregate MPC	0.328	-	0.355	0.320	-	0.358
Consumption response to MP shock	-0.18%	-	-0.19%	-0.39%	-	-0.44%
Earnings Gini response to MP shock	0.00%	-	0.16%		-	0.53%

Notes: The table shows the MPC, the earnings weight in two regimes (1) and (4), the relative response to monetary policy in two regimes (2), (4) and the weight adjusted by the relative response to monetary policy shocks in two regimes (3), (5). The columns (3) and (5) are obtained by multiplying columns (1) and (2), and columns (4) and (5), respectively. The aggregate MPC is calculated using the different weighting schemes numbered (1), (3), (4) or (5), which come from earnings or earnings adjusted with the heterogeneous exposure to the monetary policy shock by regime. The last two rows show the response of consumption and inequality to a contractionary monetary policy shock of one standard deviation (3.2 bps). The response is conditioned on the different weighting schemes and the size of the monetary policy shock. Individuals' response to the monetary policy shocks from 2006M2-2024M12 and MPCs in 2021 and 2024.

by the shock when inflation is high. The second is that the high-income groups are much less affected in the high-inflation environment. In fact, the top 1% are twice as affected by a monetary shock. The correlation between the exposure to the shock and the MPC is consequently stronger in the high-inflation environment. More details on the covariation are shown in Figures F.3 and F.4 in the appendix.

The earnings heterogeneity channel accounts for 11% of the response in aggregate consumption during a high-inflation regime and 8% of the response during a low-inflation regime. To assess the effect fully however, we need to account for the average impact of a monetary policy shock. A one standard deviation monetary policy shock is much stronger in a high-inflation regime as it has an impact of -1.2% relative to -0.5% on

average earnings; see Table D.2. Given the dominant effect of monetary policy when inflation is high, the total effect of monetary policy on inequality is even stronger in this regime. A one standard deviation monetary policy shock has more than a two times larger effect on consumption in a high-inflation regime than in a low-inflation regime. To assess the impact on inequality, we simulate our microdata and find that the effect of monetary policy on earnings inequality is also stronger during a high-inflation regime.

6 Conclusions

Using near-real-time high-frequency administrative data, we shed new light on the earnings heterogeneity channel of monetary policy. We document that monetary policy shocks substantially affect earnings inequality, with most of the effect originating from the extensive margin. Turning to the aggregate implications, we show that the earnings heterogeneity channel amplifies aggregate fluctuations and increases inequality. We assign 7.4% of the impact of monetary policy on aggregate consumption to the earnings heterogeneity channel, where the amplification works more strongly through transitions into and out of employment.

We also document that monetary policy shocks have a stronger quantitative impact on earnings inequality in a high-inflation regime, which also leads to somewhat stronger aggregate amplifications. Our findings also raise the concern that taming high inflation disproportionately affects low-income workers, as they are more exposed to monetary policy shocks during periods of high inflation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly, Claude and ChatGPT in order to help with proofreading. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendices

A Estonian Economy: Aggregate Dynamics

Estonia has a dynamic economy where the growth rate and inflation have been larger than on average in the euro area, but with more pronounced ups and downs, see Figure A.1. GDP volatility is high due to strong exposure to foreign shocks and limited automatic fiscal stabilisers. The country experienced a vast decline during the Great Recession, followed by a fast recovery and then solid growth for a decade until the Russo-Ukrainian war led to another substantial interruption of growth.

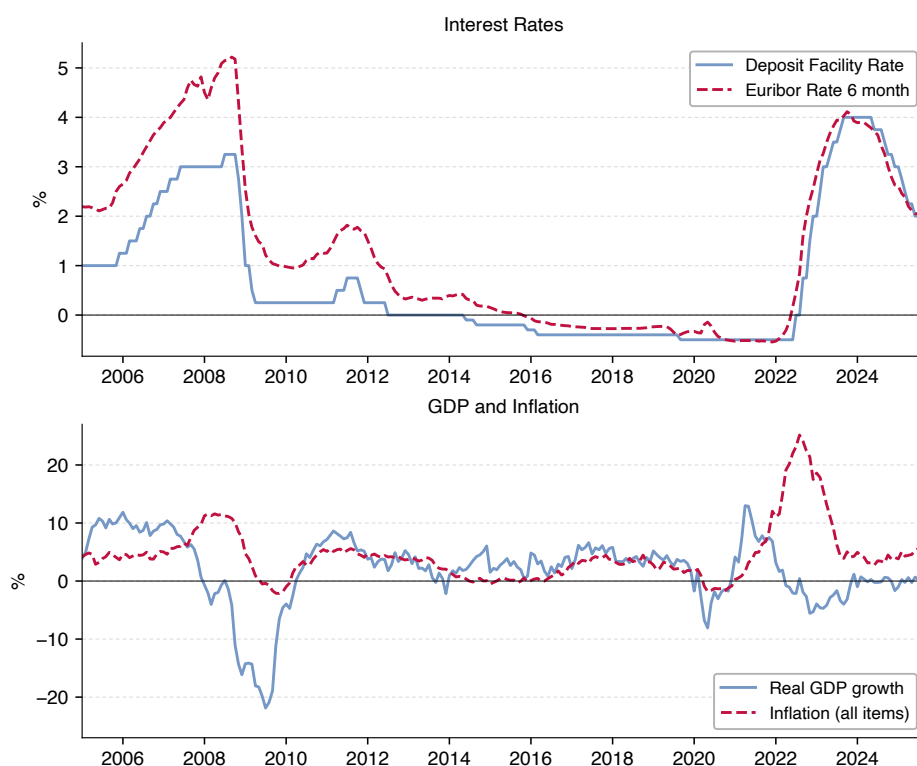


Figure A.1: Aggregate time series, 2005M1-2025M09. Upper chart: Deposit Facility Rate & the 6 month Euribor interest rate. Lower chart: real GDP growth (interpolated at monthly frequency) & inflation (HICP all items).

B Data

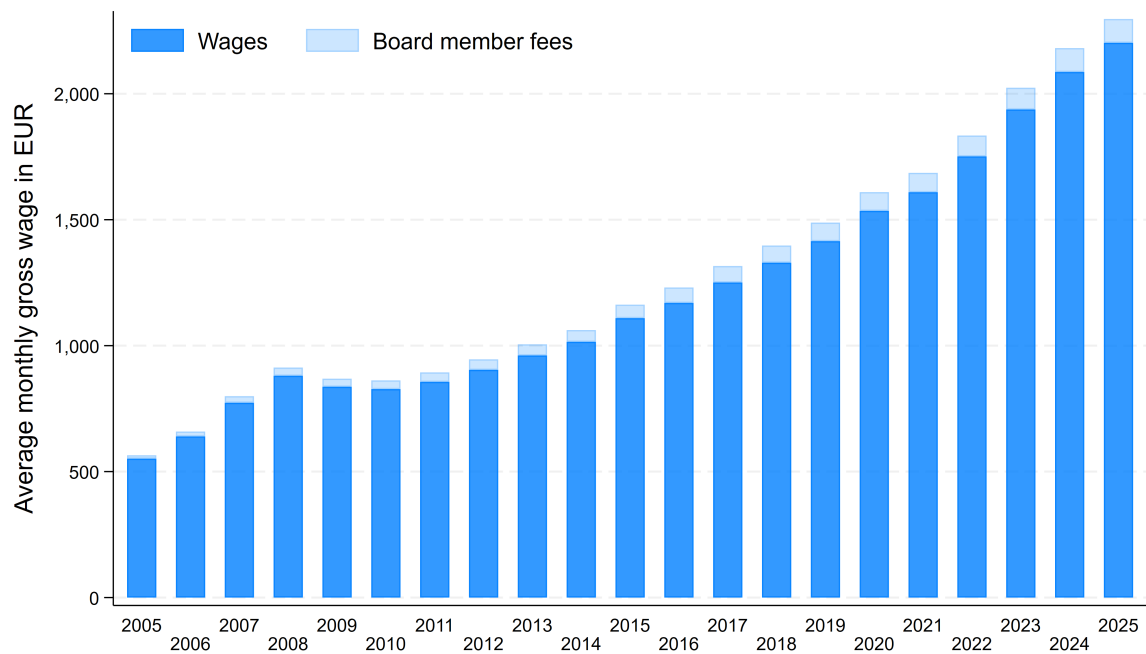


Figure B.1: Average labour income by type in 2005-2025M12.

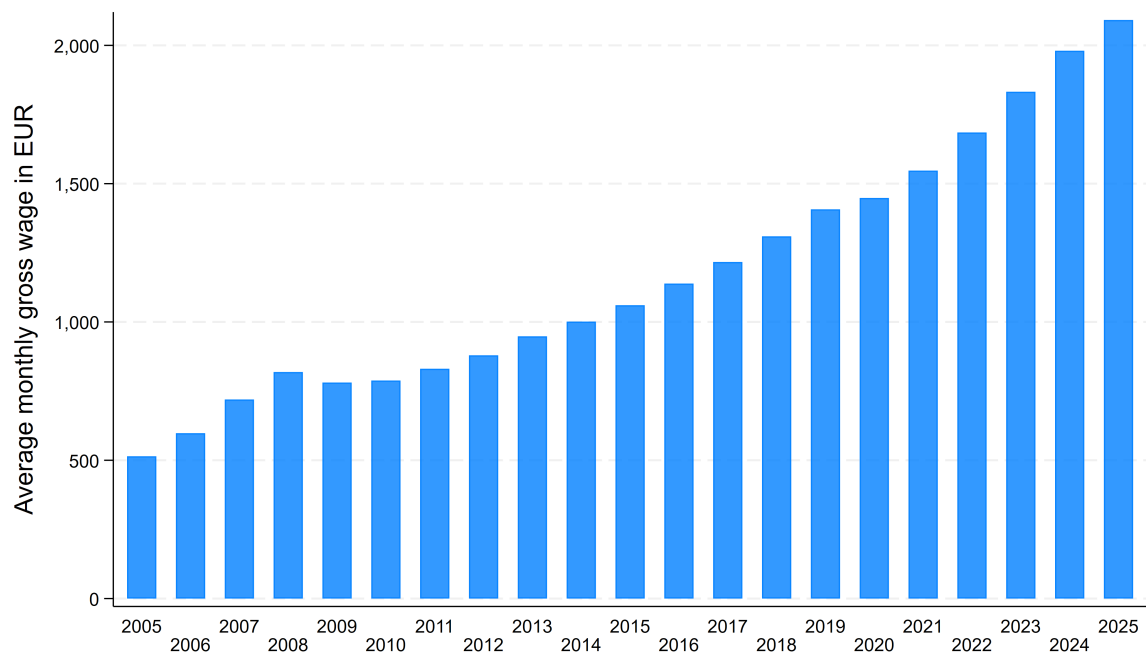


Figure B.2: Average labour income by Statistics Estonia in 2005-2025M12.

Table B.1: Earnings descriptives by permanent income group, 2006M2-2024M12

Income group from $t - 2$ to $t - 13$	(1) Mean earnings in euros	(2) Mid-point average earnings growth from $t - 1$ to $t + 12$	(3) Intensive margin growth from $t - 1$ to $t + 12$	(4) Entry rate from $t - 1$ to $t + 12$	(5) Exit rate from $t - 1$ to $t + 12$	(6) Number of observations
[0 – 10)	414.3	0.149	0.191	0.167	0.119	3,091,681
[10 – 20)	489.4	0.004	0.118	0.108	0.138	3,084,347
[20 – 30)	536.0	-0.070	0.087	0.054	0.120	3,116,124
[30 – 40)	665.5	-0.082	0.056	0.044	0.106	3,093,990
[40 – 50)	802.9	-0.092	0.038	0.032	0.094	3,097,273
[50 – 60)	954.5	-0.093	0.029	0.024	0.082	3,097,695
[60 – 70)	1132.7	-0.095	0.024	0.018	0.076	3,096,372
[70 – 80)	1368.4	-0.094	0.018	0.015	0.070	3,094,434
[80 – 90)	1763.6	-0.101	0.003	0.014	0.066	3,071,627
[90 – 99)	3002.9	-0.123	-0.031	0.012	0.060	2,778,843
[99 – 99.9)	7553.1	-0.169	-0.093	0.009	0.051	1,070,715
[99.9 – 100]	23125.5	-0.176	-0.119	0.009	0.043	124,891
All sample	1260.1	-0.060	0.038	0.049	0.093	31,817,992

Notes: Earnings are deflated by HICP in 2015 prices. The number of observations refers to the observations in the second, fourth and fifth columns. The number of observations for the third column, intensive margin, is smaller and corresponds to observations where labour income was reported at time $t - 1$ and $t + 12$. The fraction of participants, intensive margin, is reported in Table D.1.

C External validation: GDP Betas

Figures C.1 and C.2 show the GDP betas conditional on gender and age. The results are similar to that of the US. The labour income of men is more sensitive to aggregate growth than the one of women. Similarly to the US, we find that for most of the earnings distribution, the youngest workers are the most sensitive to aggregate growth. Lastly, we repeat the baseline estimates reported in the main text, but with heteroscedasticity robust standard errors, following the original paper by Guvenen et al. (2017). The results are shown in Figure C.3 and demonstrate substantially narrower confidence bounds than in the case of our baseline with two-way clustered standard errors.

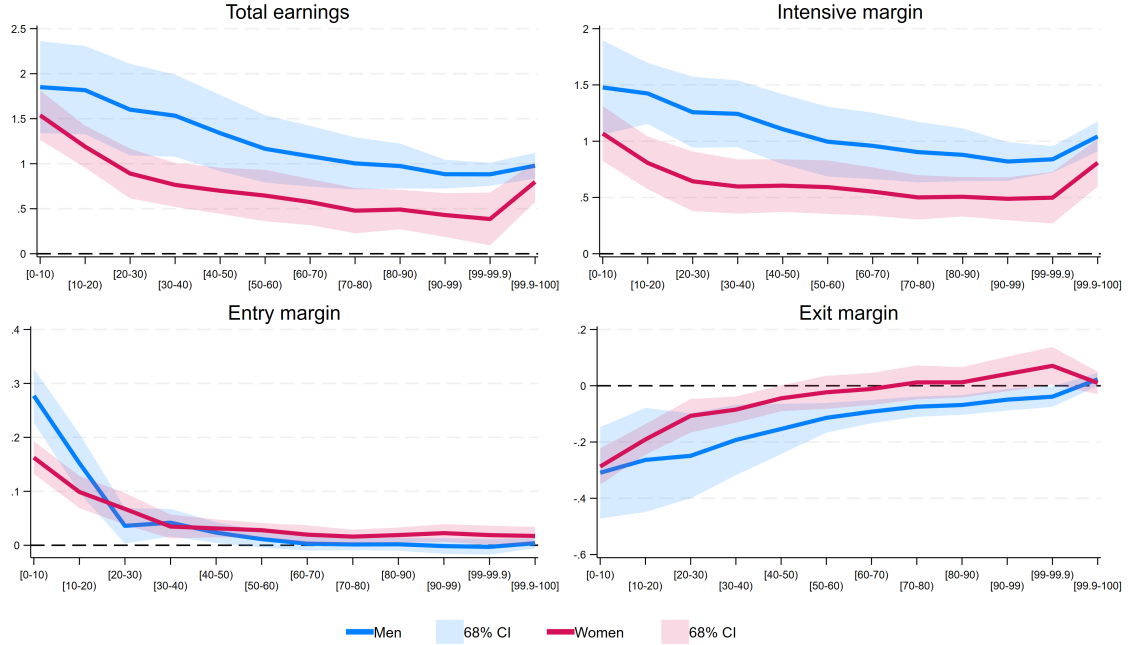


Figure C.1: GDP betas by earnings groups & gender, yearly data, 2006-2024.

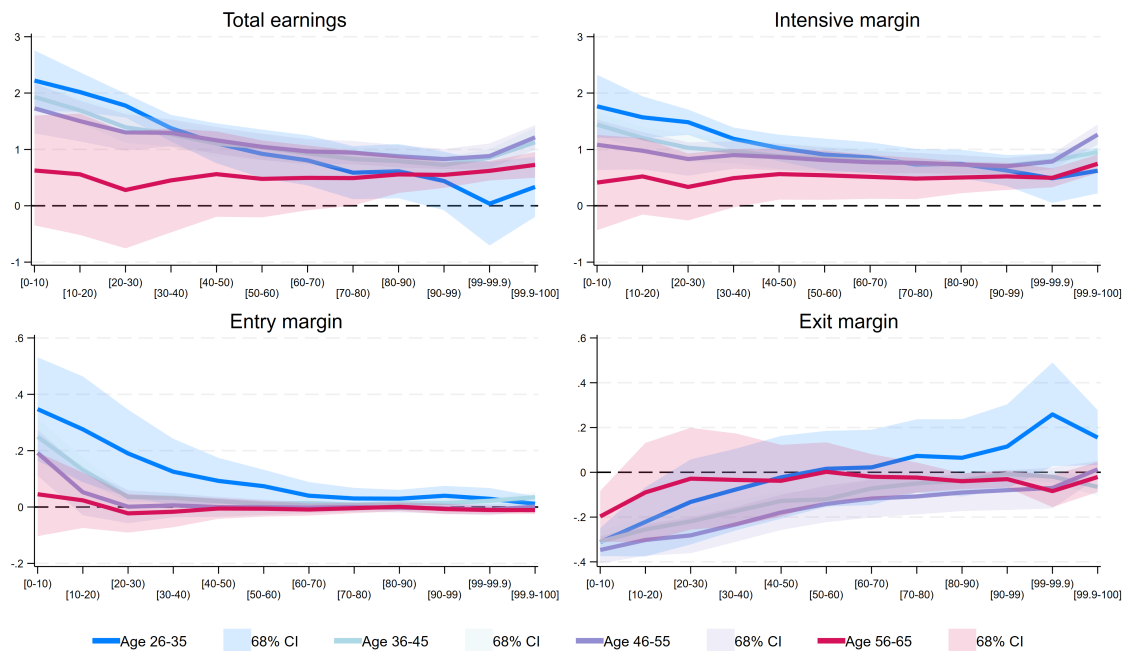


Figure C.2: GDP betas by earnings & age groups, yearly data, 2006-2024.

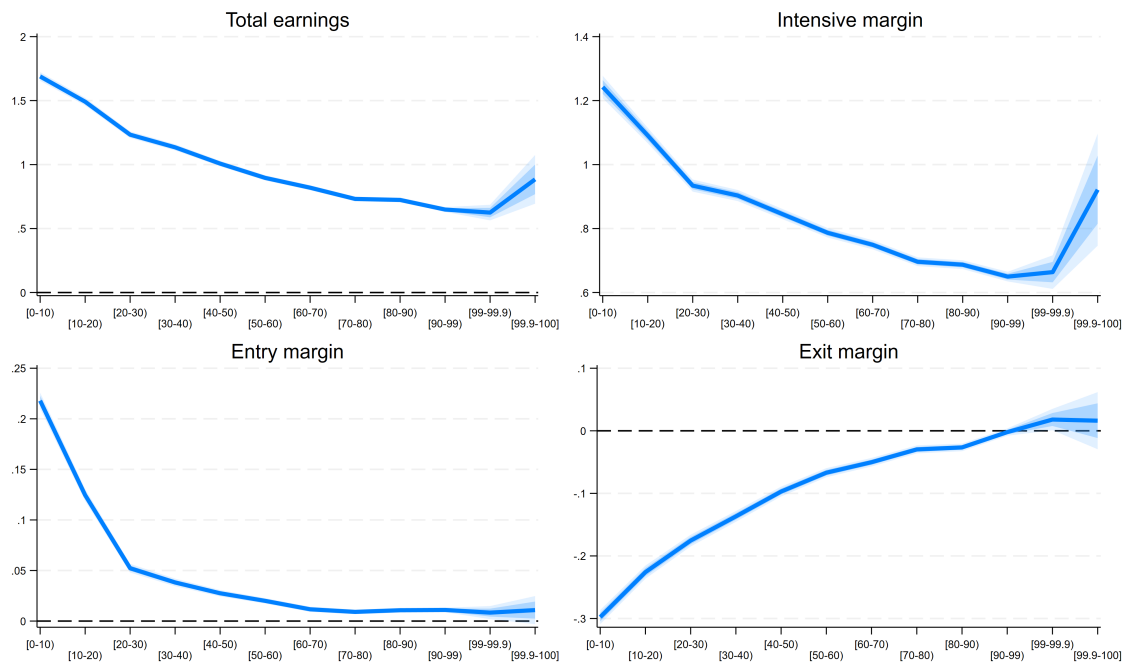


Figure C.3: GDP betas by earnings groups, heteroscedasticity robust standard errors, yearly data, 2006-2024.

D Robustness Tests

D.1 Benchmark: Yearly versus Monthly Frequency

Figure D.1 shows the total impact, intensive and extensive margins for an annual frequency. A comparison of the decomposition at monthly and annual frequencies, as shown in Figure 4, highlights important differences. Specifically, it shows that the annual data substantially overestimates the intensive margin, as it misses a lot of within-year transitions into and out of employment. Thus, a majority of the monetary policy effect originates by construction from the intensive margin for the annual data. Consequently, the intensive margin at monthly frequency only accounts for half of the adjustment relative to annual data, as shown in Table D.1.

Table D.1 also demonstrates the economic size of the estimated effects. The shocks at monthly and yearly frequencies are not directly comparable because the expansionary and contractionary shocks cancel each other out at the yearly frequency. By standard deviation terms, the yearly shock is three times larger than the monthly shock (9.7 basis points versus 3.2 basis points). The specification at yearly frequency differs also from a monthly frequency by the impact horizon h . At a monthly frequency, the effect of shock in month t by month $t + 12$ is studied; at a yearly frequency, the effect of shock throughout the year t by the next year is studied, which incorporates the impact of shocks by up to month 23.

A one standard deviation contractionary shock reduces real wage growth by month 12, the intensive margin, by 5.2% in our yearly data and by 0.3% in the monthly data. This estimate aligns with findings that monetary policy has a stronger pass-through in Estonia than in other euro area countries. For example, Hubert and Savignac (2024) find that a one standard deviation shock reduces total wage income, intensive margin, by 0.44% in French data at yearly frequency. Broer et al. (2022) estimate a 0.45% fall for the aggregate wage growth, intensive margin, with German monthly data. The number aligns with Almgren et al. (2022), who find the peak and cumulative effect of monetary policy in Estonia to be 6-7 times larger than in France and 2-3 times larger than in Germany. Amberg et al. (2022) finds that the earnings of the first income decile decline by 2% due to 25bps shock in Sweden, which corresponds to roughly 3 times lower responsiveness in their data compared to ours (the standard deviation of their monetary policy shock is 33bps, and a one standard deviation shock corresponds to 2.6% decline in the earnings of the first income decile).

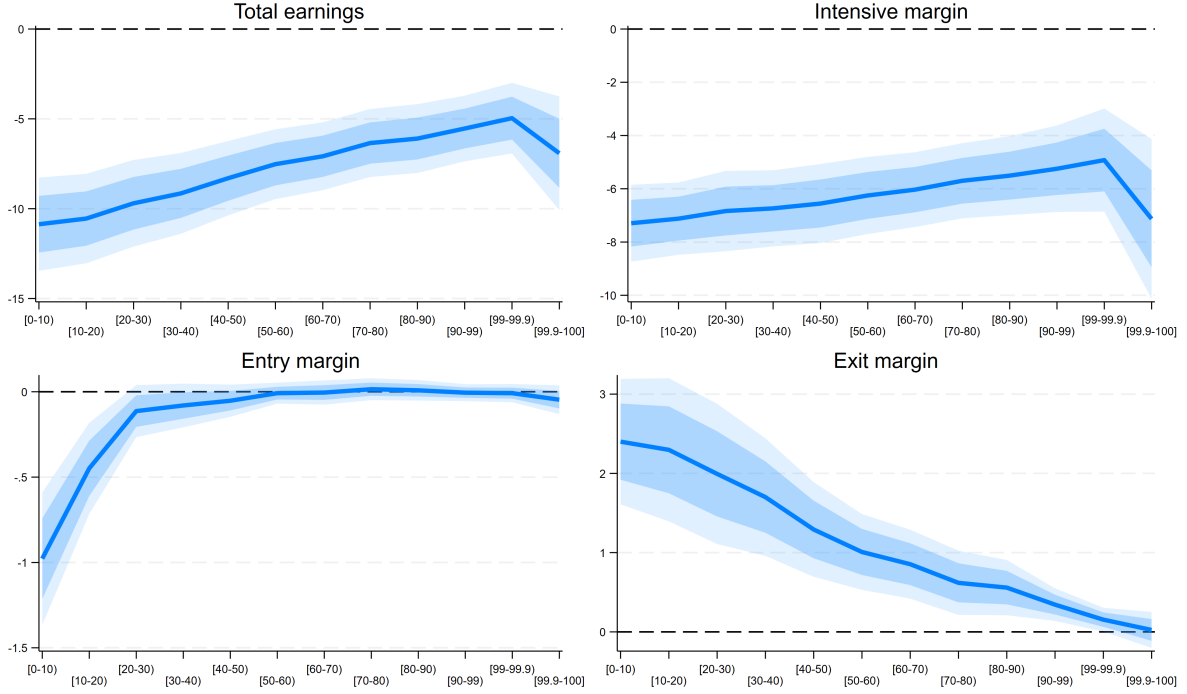


Figure D.1: The effect of a monetary policy shock (1SD) on earnings growth (%) by earnings groups, yearly data, 2007-2024.

Notes: The effect on total earnings, the intensive margin, the entry margin and the exit margin are shown. The specification at yearly frequency studies the effect of monetary policy shock in year t on the growth of dependent variables over $y_{i,t+1}$ and $y_{i,t-1}$, and controls for macro variables at $\Delta X_{i,t-1}$, i.e. $\Delta y_{i,t+1} = \sum_{g=1}^{12} G_{i,t-2,g} \times (\alpha_g^h + \beta_g^h \Delta i_t + \gamma_g^h \Delta X_{t-1}) + \epsilon_{i,t+h}$. Each value in a figure represents estimates of $(\beta_g^h * 0.097) \times 100$ for horizon $h = 1$, indicating the effect of 1SD monetary policy shock on the accumulated earnings growth of group g by year 1. The sample 2007-2024 refers to the effective sample, i.e. dates of monetary policy shocks we are studying; the earnings data from 2005-2025 are used to obtain these estimates. The standard errors are clustered by individuals and years, shaded areas represent 90 and 68% CIs.

Table D.1: The effect of a monetary policy shock (1SD) on earnings growth (%) by earnings groups, at monthly and yearly frequency

	Monthly frequency 2006M2-2024M12			Yearly frequency 2007-2024		
	(1) Total earnings	(2) Partici- pation	(3) Contribution of intensive margin	(4) Total earnings	(4) Partici- pation	(5) Contribution of intensive margin
[0 – 10)	-1.3**	0.271	-0.2	-10.4***	0.507	-3.7***
[10 – 20)	-1.6**	0.526	-0.3**	-10.4***	0.700	-5.0***
[20 – 30)	-1.4**	0.708	-0.4**	-9.7***	0.810	-5.5***
[30 – 40)	-1.3**	0.753	-0.4*	-9.2***	0.842	-5.6***
[40 – 50)	-1.3*	0.800	-0.5*	-8.4***	0.870	-5.7***
[50 – 60)	-1.1*	0.832	-0.4	-7.6***	0.890	-5.5***
[60 – 70)	-1.0	0.851	-0.4	-7.1***	0.902	-5.4***
[70 – 80)	-0.7	0.865	-0.2	-6.4***	0.911	-5.2***
[80 – 90)	-0.7	0.875	-0.2	-6.2***	0.919	-5.0***
[90 – 99)	-0.4	0.887	-0.1	-5.6***	0.928	-4.9***
[99 – 99.9)	-0.2	0.910	-0.1	-4.9***	0.937	-4.6***
[99.9 – 100]	-0.9	0.921	-0.6	-6.8***	0.941	-6.7***
All sample	-1.1*	0.737	-0.3	-8.1***	0.829	-5.2***

Notes: Total earnings indicates the effect of 1SD monetary policy shock on the accumulated earnings growth of group g by month 12 or by year one. The contribution of the intensive margin is derived by multiplying the effect of monetary policy by this margin with the fraction of workers' participating in both periods $y_{i,t+h}$ and $y_{i,t-1}$. 1SD corresponds to a 3.2 basis point shock at monthly frequency and a 9.7 basis point shock at yearly frequency. The sample 2006M2-2024M12 or 2007-2024 refers to the effective sample, i.e. dates of monetary policy shocks we are studying; the earnings data from 2005M1-2025M12 or 2005-2025 are used to obtain these estimates. The standard errors are clustered by individuals and dates, ***, ** and * refer to statistical significance at 1, 5 and 10% level.

D.2 Extensions: Horizon Length, Gender and Age

We present further results for our benchmark specification here. We show estimates, where we condition the impact on gender and age.

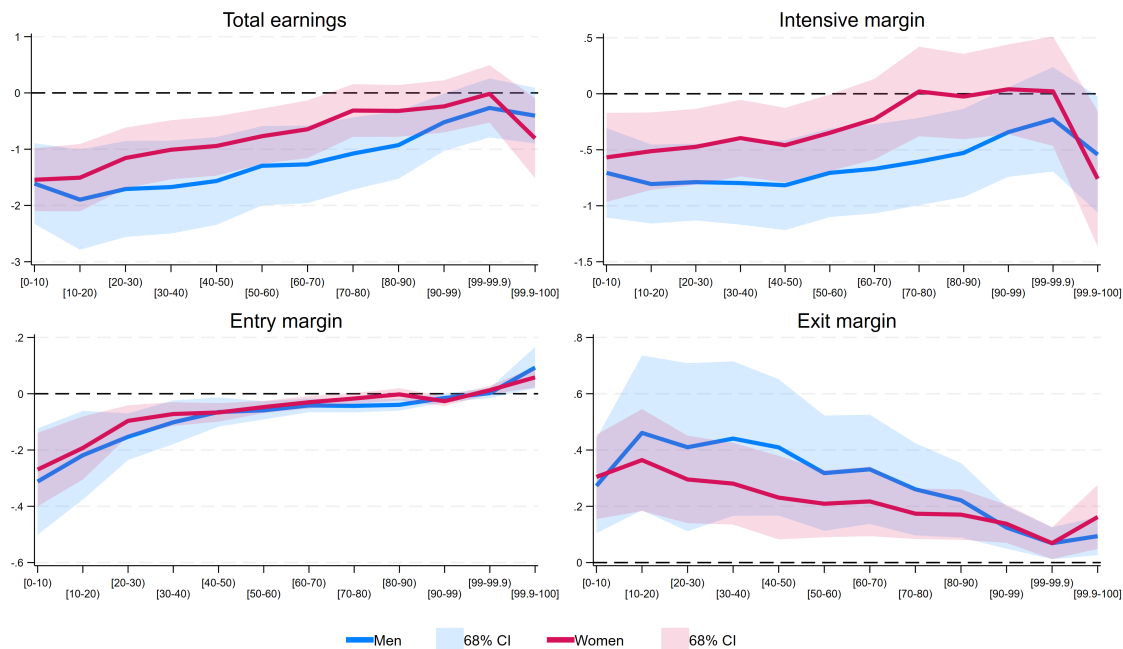


Figure D.2: The effect of monetary policy shock on earnings by earnings groups and gender, 2006M2-2024M12.

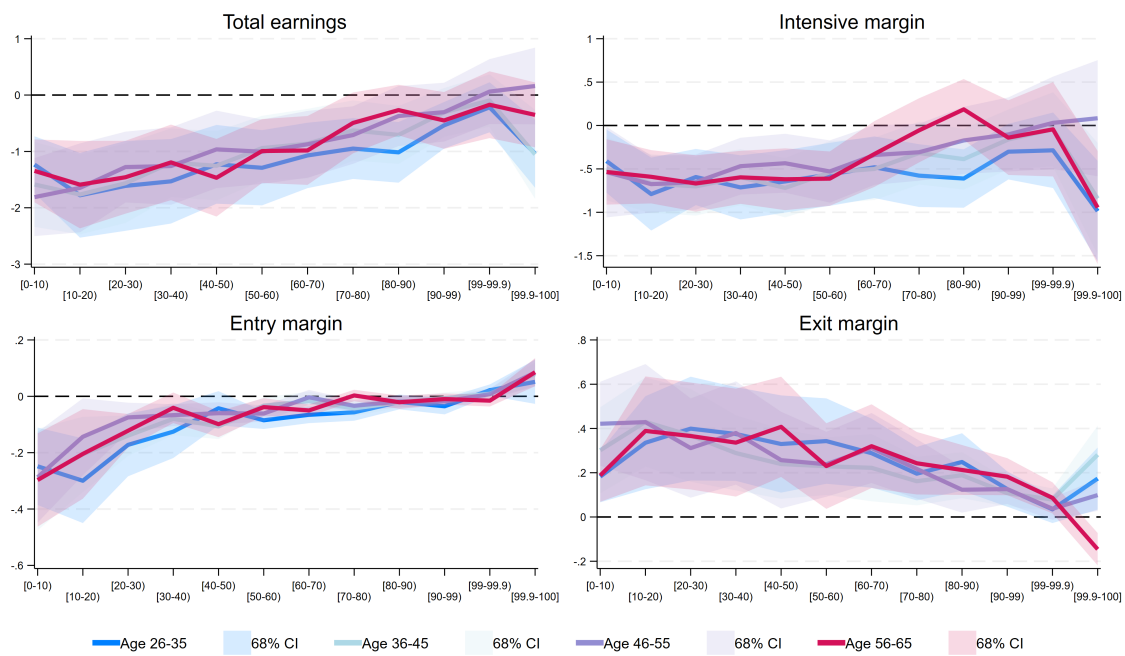


Figure D.3: The effect of monetary policy shock on earnings by earnings groups and age, 2006M2-2024M12.

D.3 Instrumental variable approach

Here we describe the methodology for the alternative monetary policy identification using the instrumental variable method. We follow the approach of Almgren et al. (2022), who instrument the policy rate EONIA with surprises in the three-month OIS rate in a narrow window around ECB announcements.⁴⁰ The identifying assumption is that no other information is released during the short window that is related to the ECB’s policy decision. The first-stage regression of the 2SLS approach is given as:

$$i_t = c + \rho Z_t + \delta Z_{t-1} + \zeta i_{t-1} + \sigma \Delta X_{t-1} + M_m + \epsilon_t, \quad (\text{D.1})$$

where i_t denotes EONIA, Z_t is the instrument of surprise change in three-month OIS rate around the monetary policy window, and ΔX_{t-1} captures the same macro controls of GDP growth in Estonia, in the Euro area and inflation in Estonia, as in our baseline specification. M_m are monthly dummies to control for seasonality.

In the second stage, we then use the fitted values $\hat{i}_t$ to estimate the effect of a monetary policy shock on labour income growth by income groups. Besides replacing the shock series of Jarociński and Karadi (2020), it is the same specification as for our baseline estimation:

$$\Delta y_{i,t+h} = \sum_{g=1}^{12} G_{i,t-2,g} \times (\alpha_g^h + \beta_g^h \hat{i}_t + \gamma_g^h \Delta X_{t-1}) + M_m + \epsilon_{i,t+h}, \quad (\text{D.2})$$

where the dependent variable $\Delta y_{i,t+h} = (y_{i,t+h} - y_{i,t-1}) / ((y_{i,t+h} + y_{i,t-1})/2)$ is the mid-point average growth of labour income of individual i in month $t + h$.

The results show very similar distributional effects of monetary policy by this approach, β_g^h , as displayed in Figure 6. The effect of a one standard deviation increase in EONIA (1.6bps) is shown. The average effect of a one standard deviation increase in EONIA over all the earnings bins is -1.4%. Note that we observe a considerably stronger U-shape, driven by the top 0.1% income earners.

⁴⁰The surprises in the three-month OIS rate around the monetary policy event window are obtained from Altavilla et al. (2019). The collection of EONIA time series was discontinued at the end of 2021. We continue this time series with the Euro area short-term rate, ESTR.

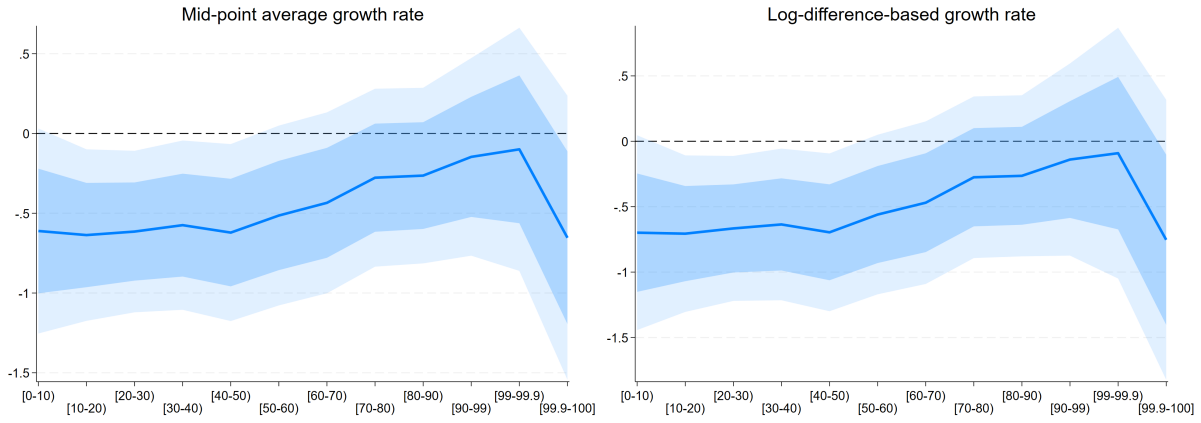


Figure D.4: Alternative growth rate measure, intensive margin, 2006M2-2024M12

D.4 Additional robustness

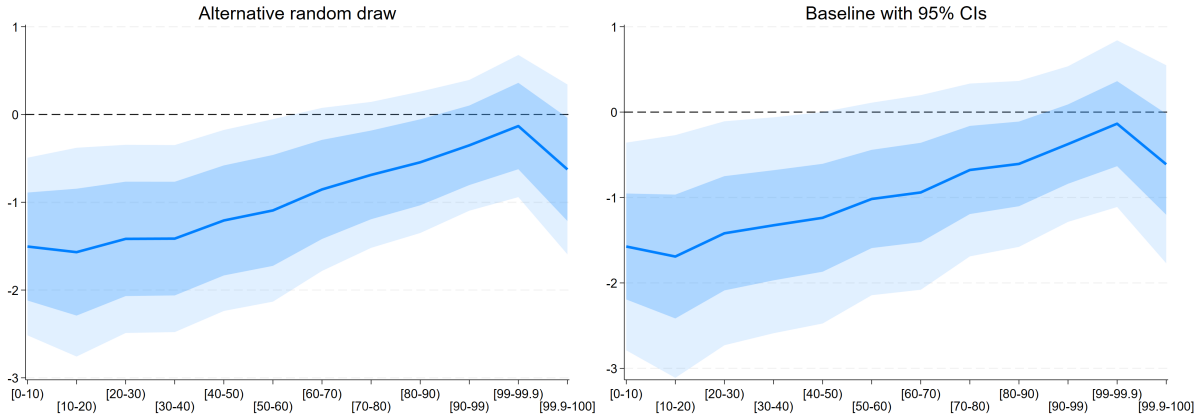


Figure D.5: The effect of monetary policy shock on earnings by earnings groups, alternative random draw of individuals and alternative confidence bounds, 2006M2-2024M12.

D.5 Inflation Dependent Specification

Here, we present further results for our inflation dependent specification.

Table D.2: The effect of 1SD monetary policy shock on earnings by earnings groups and inflation regime, 2006M2-2024M12

	(1)	Low inflation		(4)	High inflation	
	Total earnings	Participation	Contribution of intensive margin	Total earnings	Participation	Contribution of intensive margin
[0 – 10)	-0.72*	0.270	-0.20	-1.46**	0.280	-0.19**
[10 – 20)	-0.74*	0.524	-0.38	-1.92**	0.536	-0.36*
[20 – 30)	-0.68*	0.712	-0.40	-1.70**	0.689	-0.50*
[30 – 40)	-0.67*	0.756	-0.45	-1.58*	0.740	-0.38
[40 – 50)	-0.66	0.802	-0.43	-1.44*	0.786	-0.52
[50 – 60)	-0.58	0.834	-0.36	-1.16	0.824	-0.42
[60 – 70)	-0.46	0.853	-0.33	-1.12	0.844	-0.27
[70 – 80)	-0.36	0.866	-0.23	-0.70	0.860	-0.06
[80 – 90)	-0.34	0.875	-0.19	-0.55	0.872	-0.07
[90 – 99)	-0.21	0.887	-0.08	-0.25	0.887	0.05
[99 – 99.9)	0.003	0.909	0.02	-0.11	0.916	0.08
[99.9 – 100]	-0.06	0.919	-0.05	-1.37	0.930	-0.97
All sample	-0.54	0.738	-0.30	-1.23*	0.732	-0.27

Notes: Earnings are deflated by HICP in 2015 prices. 1SD corresponds to a 3.2 basis point shock. ***, ** and * refer to statistical significance at 1, 5 and 10% level based on two-way clustered standard errors by individuals and dates.

E Derivation of the Aggregate MPC

We use a simple model based on Patterson (2023) to derive the aggregate MPC for a monetary policy shock. Our focus is on the earnings channel of monetary policy. The household j has the following consumption function $C_j = C_j(E_j(Y(mp)), \theta_j)$. Earnings is a function of output, which is again a function of monetary policy. We assume that there is no capital or profits so that we have $Y = \sum_j E_j$.

The starting point is how consumption responds to a monetary policy shock:

$$\frac{dC}{dmp} = \sum_j \frac{dC_j}{dE_j} \frac{dE_j}{dmp} = \left(\sum_j \frac{E_j}{Y} \frac{dC_j}{dE_j} \gamma_j^{mp} \right) \frac{dY}{dmp}. \quad (\text{E.1})$$

where $\gamma_j^{mp} = \frac{Y}{E_j} \frac{dE_j}{dmp} \frac{dmp}{dY}$. We can then use the earnings-weighted covariance definition, which is

$$\text{cov} \left(\frac{dC_j}{dE_j}, \gamma_j^{mp} \right) = \left(\sum_j \frac{dC_j}{dE_j} \gamma_j^{mp} \right) - \left(\sum_j \frac{E_j}{Y} \frac{dC_j}{dE_j} \right) \left(\sum_j \frac{E_j}{Y} \gamma_j^{mp} \right). \quad (\text{E.2})$$

Note that we have

$$\left(\sum_j \frac{E_j}{Y} \gamma_j^{mp} \right) = \left(\sum_j \frac{E_j}{Y} \frac{Y}{E_j} \frac{dE_j}{dmp} \frac{dmp}{dY} \right) = \frac{dmp}{dY} \sum_j \frac{dE_j}{dmp} = 1. \quad (\text{E.3})$$

where we use that $\sum_j E_j = Y$. As a next step, we can now use this expression and write

$$\frac{dC}{dmp} = \left[\left(\sum_j \frac{E_j}{Y} \frac{dC_j}{dE_j} \right) + \text{cov} \left(\frac{dC_j}{dE_j}, \gamma_j^{mp} \right) \right] \frac{dY}{dmp}. \quad (\text{E.4})$$

The aggregate MPC for a monetary policy shock is given as

$$\text{Aggregate MPC}|_{mp} = \frac{dC/dmp}{dY/dmp} = \left(\sum_j \frac{E_j}{Y} \frac{dC_j}{dE_j} \right) + \text{cov} \left(\frac{dC_j}{dE_j}, \gamma_j^{mp} \right), \quad (\text{E.5})$$

which is the searched equation.

F Aggregate Implications

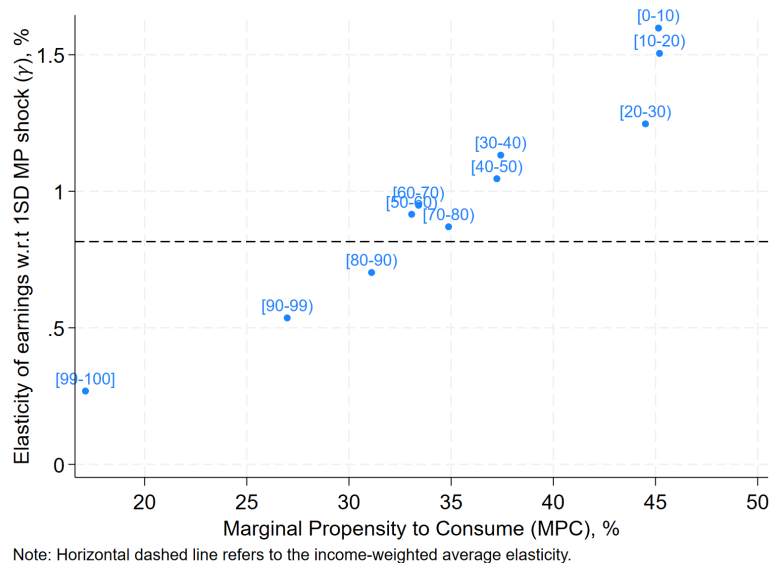


Figure F.1: MPCs and earnings elasticity with respect to monetary policy by earnings groups, simulation on HFCS data from 2021 and 2024.

Notes: Figure demonstrates elasticity for accommodative monetary policy shock. Income-weighted elasticity reported in figure, 0.8%, is lower than the average elasticity reported in Table D.1), 1.1%, because responsive low-wage workers account for a smaller share in total earnings after introducing earnings weights.

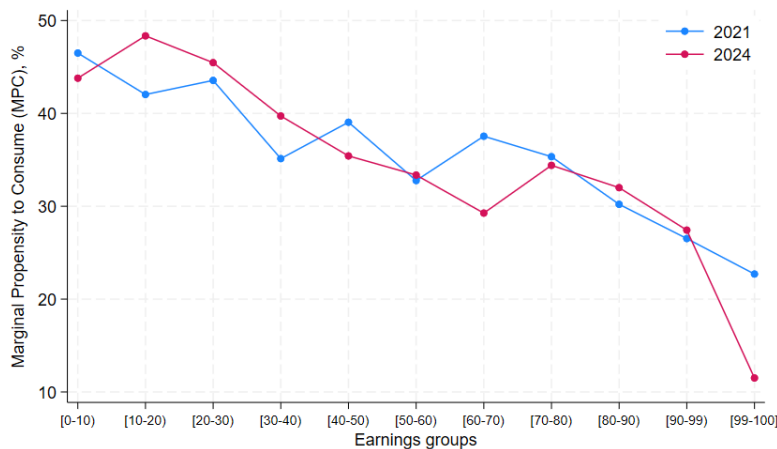


Figure F.2: MPC dynamics by earnings groups, before and after the inflation surge.

Notes: Figure demonstrates the average MPCs by earnings groups collected before the inflation surge, between January and August 2021, and after the inflation surge, between February and June 2024.

Table F.1: The MPCs and monetary policy effects by household earnings groups and response margin

Earnings percentile	Intensive margin			Extensive margin		
	(1) Earnings weight	(2) Relative response	(3) MP weight	(4) Earnings weight	(5) Relative response	(6) MP weight
[0 – 10)	0.014	0.851	0.012	0.014	2.681	0.036
[10 – 20)	0.032	1.368	0.044	0.032	2.104	0.067
[20 – 30)	0.047	1.372	0.064	0.047	1.602	0.075
[30 – 40)	0.060	1.256	0.076	0.060	1.448	0.087
[40 – 50)	0.072	1.179	0.085	0.072	1.348	0.098
[50 – 60)	0.087	1.053	0.091	0.087	1.191	0.103
[60 – 70)	0.105	1.219	0.129	0.105	1.117	0.118
[70 – 80)	0.130	1.101	0.143	0.130	1.041	0.135
[80 – 90)	0.169	0.926	0.156	0.169	0.812	0.137
[90 – 99)	0.230	0.764	0.176	0.230	0.568	0.131
[99 – 100]	0.054	0.459	0.025	0.054	0.239	0.013
Weighted Aggregate MPC	0.324	-	0.343	0.324	-	0.353
Consumption response to MP shock	-0.105%	-	-0.112%	-0.246%	-	-0.268%
Earnings Gini response to MP shock	0.00%	-	0.07%		-	0.28%

Notes: The table shows the MPC, the income weights (1) and (4), the relative response to monetary policy by the two margins (2), (5) and the weight adjusted by the relative response to monetary policy shocks by the two margins (3) and (6). The columns (3) and (6) are obtained by multiplying columns (1) and (2), and columns (3) and (4), respectively. The aggregate MPC is calculated using the different weighting schemes numbered (1), (3), (4) or (6), which come from earnings or earnings adjusted with the heterogeneous exposure to the monetary policy shock by the two margins. The last two rows show the response of consumption and inequality to a contractionary monetary policy shock of one standard deviation (3.2 bps). The response is conditioned on the different weighting schemes and the size of the monetary policy shock. Individuals' response to the monetary policy shocks from 2006M2-2024M12 and MPCs in 2021 and 2024.

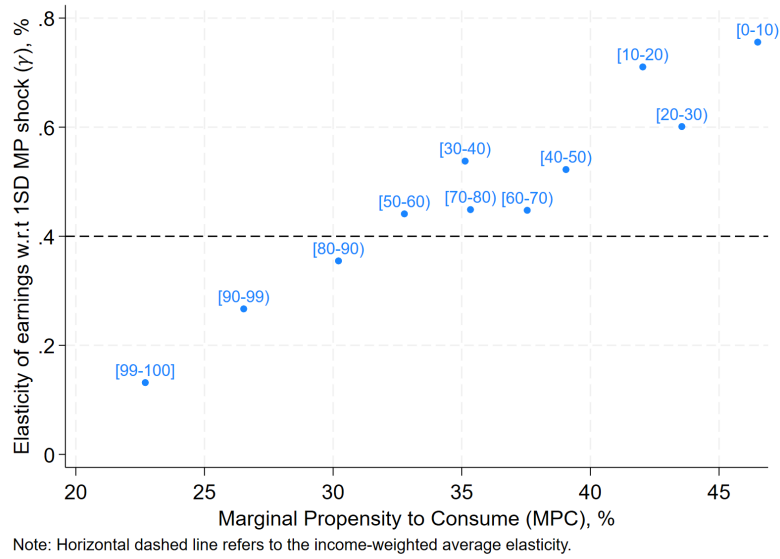


Figure F.3: MPCs and earnings elasticity with respect to monetary policy by earnings groups during low-inflation regime, simulation on HFCS data from 2021.

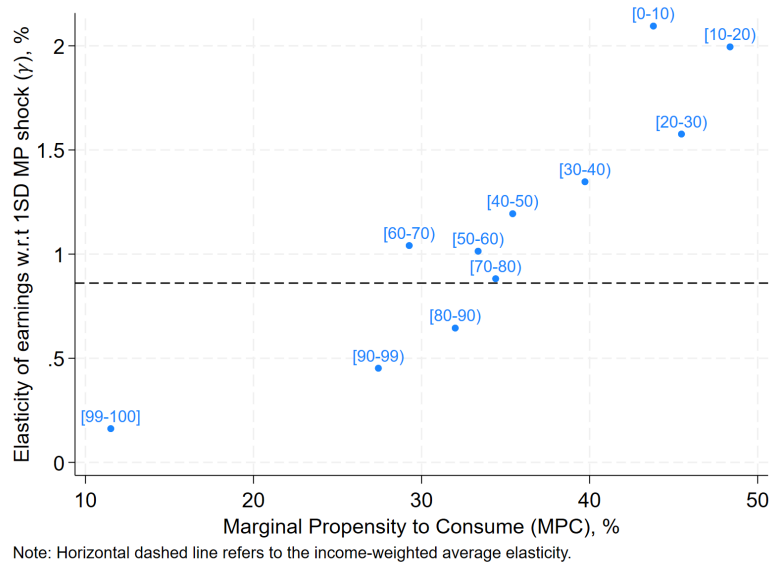


Figure F.4: MPCs and labour earnings elasticity with respect to monetary policy by earnings groups during high-inflation regime, simulation on HFCS data from 2024.